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Chapter 14. Manuscript frames 786–820. Catalogue snapshot: 5 October 2026.

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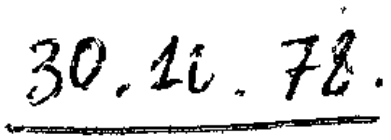
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Chapter 14

GELFAND-P000786

[source number: 4]

30.10.78. Gelfand Seminar



Uncertain reading. The month looks more like 10. A break in the second digit leaves 11 possible.

Original source details: GELFAND-P000786

Kruskal

Surreal numbers

there are 2 books: John Conway — (“combinatorial games”).

{Knuth D. “[crossed out: ...] — novel”}

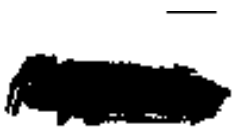


Undeciphered passage. Dense hatching obscures the entry before “— novel”.

Original source details: GELFAND-P000786

Def.

Number = map from a well-ordered set to a set of 2 elements ↑ and ↓.



Undeciphered passage. A heavily inked horizontal mark lies below the wavy line. No letters can be distinguished.

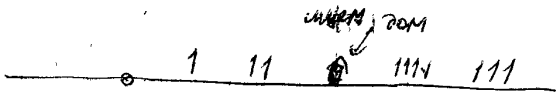
Original source details: GELFAND-P000786

$x \neq y$ — compare the arrows (lexicographically).

$$\widehat{\uparrow} = \uparrow\uparrow\uparrow \ldots = \omega$$

$\widehat{\ } \leftrightarrow$ (periodic)

$\widehat{\uparrow\downarrow}$ $\widehat{\downarrow}$

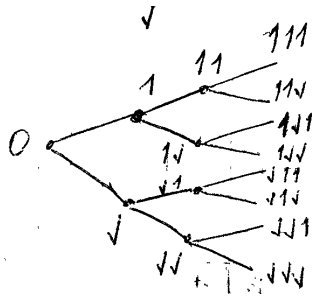


Original source details: GELFAND-P000786

Diagram and typed description.

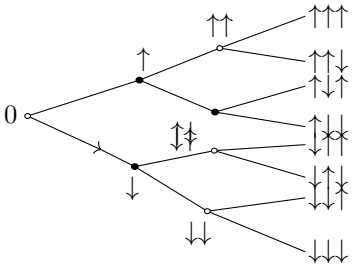


The crossed-out word before “house” is illegible.

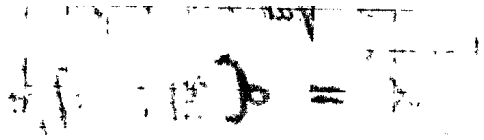


Original source details: GELFAND-P000786

Diagram and typed description.



$\ldots\} = \ldots$



Original source details: GELFAND-P000786

Undeciphered passage. A right brace and an equals sign are visible under the number line. The other faint strokes do not establish the expression.

$$x \underset{\text{earlier (earlier)}}{\longleftarrow} y$$

(i.e. x is on the path from 0 to y)

$y]$ — an element of the set of numbers that come before y

Truncation operation — x has fewer arrows

$y\rceil$ a number earlier than y, but greater than it

$y\rfloor$ —"— and less than it

$x < y$ is also defined

$$\{y]\} = \{y\rfloor\} \cup \{y\rceil\},$$
$$\{y\rceil\} \cap \{y\rfloor\} = \Lambda$$

Example:

Main def.:

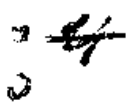
GELFAND-P000787

Thm. 1

(simple)

A, B — sets of numbers

$\forall a \in A, b \in B$; [crossed out: ...] $a < b$



Original source details: GELFAND-P000787

Uncertain reading. A heavily crossed-out sign lies between the semicolon and the inequality.

Then $\exists x: \quad a < x < b$.

and there exists a unique earliest x.

Proof: transfinite ind., we look at $x = 0$ and if we have not found x up to a certain point, we look further.

Notation:

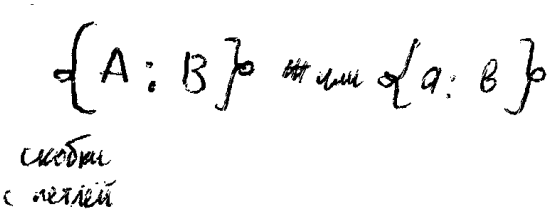
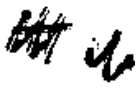


Diagram and typed description.
$\{A : B\}$ ... or $\{a : b\}$

braces with a loop

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Uncertain reading. The cancelled sign before "or" resembles Russian «и» ("and").



Original source details: GELFAND-P000787

Addition.

by transfinite induction.

Theorem 2

$$x = \{x\rfloor : x\rceil\}$$

(obvious)

Def. of addition.

$$x + y = \{x\rfloor + y,\ x + y\rfloor :\ x\rceil + y,\ x + y\rceil\}$$

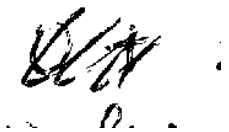
Theorem 3.

a) $0 + 0 = \{:\} = 0$.

[Below the formula:] there are no conditions, so the earliest of all

b) [crossed out: ...] $x + 0 = x$ by induction

Uncertain reading. The cancelled strokes resemble $\forall x$.



Original source details: GELFAND-P000787

Proof:

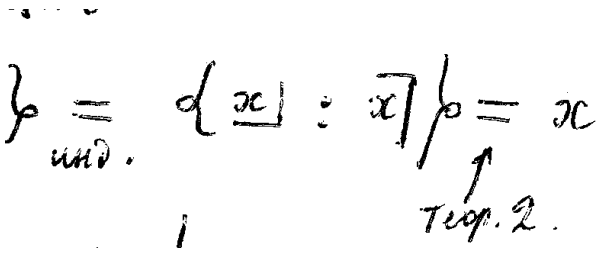


Diagram and typed description.
$x + 0 = \{x\rfloor + 0 : x\rceil + 0\}$
$\underset{\text{ind.}}{=} \{x\rfloor : x\rceil\} \underset{\text{Thm. 2.}}{=} x$

Original source details: GELFAND-P000787

In the ordinary theory there is no notion of earlier!

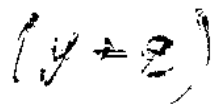
$x + y$ — the simplest number, monotone in each argument

a) monotone in x, y

b) the earliest of all such numbers.

The usual properties hold

$$x + y = y + x$$
$$(x + y) + z = x + (y\ [?]\ z)$$



Uncertain reading. The sign between y and z resembles $\geq$. The reading $+$ is also possible.

Original source details: GELFAND-P000787

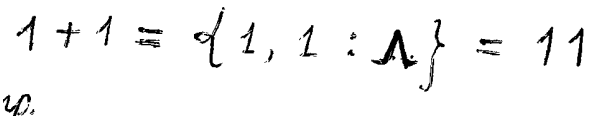
There is an inverse operation.

The opposite of x — reversal of the arrows $\quad x + \mathrm{opp}\, x = 0.$

GELFAND-P000788

[source number: 5]

$$\uparrow + \uparrow = \{1, 1 : \Lambda\} = \uparrow\uparrow$$



Uncertain reading. The initial signs on either side of the plus resemble up arrows. “1+1” is also a possible reading. The ones within the braces have feet.

Original source details: GELFAND-P000788

Ex.

$$\uparrow\downarrow + \uparrow\downarrow = ?$$
$$\hat{\uparrow} + \downarrow\downarrow\downarrow = ?$$

Multiplication.

Basic inequality of mult’n.

$$r < s, \qquad u < v$$



Uncertain reading. The sign before $< s$ is provisionally typeset as r. Its shape also admits η.

Original source details: GELFAND-P000788

Then

$$ru + sv \text{ [?] } rv + su$$



Original source details: GELFAND-P000788

Uncertain reading. The first letters in ru and rv have been retraced. The sign after sv is overwritten and resembles $>$.

xy — any of the four [?] $\Rightarrow$ 4 inequalities



Uncertain reading. The abbreviation after "four" may be «слаг.» (summands) or «оп.» (options).

Original source details: GELFAND-P000788

$$\begin{aligned} x \cdot y = \{&\lfloor x \rfloor \cdot y + x \cdot y \rfloor - \lfloor x \rfloor \cdot \lfloor y \rfloor, \\ &\lceil x \rceil \cdot y + x \cdot y \rceil - \lceil x \rceil \cdot \lceil y \rceil : \\ &\lfloor x \rfloor \cdot y + x \cdot y \rceil - \lfloor x \rfloor \cdot \lceil y \rceil, \\ &\lceil x \rceil \cdot y + x \cdot y \rfloor - \lceil x \rceil \cdot \lfloor y \rfloor\} \end{aligned}$$



Uncertain reading. A heavy cross-shaped mark stands above the comma at the end of the third line. Its purpose is unclear.

Original source details: GELFAND-P000788

The natural properties of multiplication hold.

$$x \cdot 0 = \{:\} = 0, \qquad x \cdot 1 = x$$

$$\begin{aligned} x \cdot 1 = \{&\lfloor x \rfloor \cdot 1 + x \cdot 0 - x \rfloor \cdot 0 : \\ &\lceil x \rceil \cdot 1 + x \cdot 0 - x \rceil \cdot 0\} \\ = \{&\lfloor x \rfloor + 0 - 0 : x \rceil + 0 - 0\} = x \end{aligned}$$

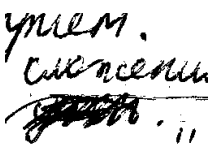
Associat., commut., distribut. hold.

Prove at home.

Advice. Technical device.

Prove that the def. [crossed out: ?] [inserted above the line with a caret:] of addition is "indifferent to brackets"

bracket indifferent



Uncertain reading. The heavily crossed-out abbreviation may read «умн.» (mult.).

Original source details: GELFAND-P000788

$$x = \{a : b\}$$
$$y = \{c : d\}$$
$$x + y = \{a + y, x + c : b + y, x + d\}$$

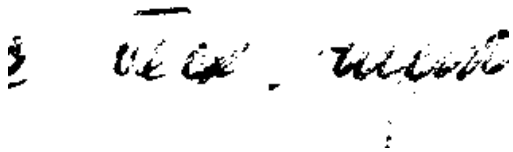
Multiplication also has this property.

Associat.

There exist inf. large and inf. small numbers.

GELFAND-P000789

$\uparrow\uparrow\uparrow \ldots = \widehat{\uparrow} = \omega$ — the simplest inf. number.



Uncertain reading. The abbreviation before «число» (“number”) provisionally reads «беск.» (inf.).

Original source details: GELFAND-P000789

$\uparrow\downarrow\downarrow \ldots = \uparrow \widehat{\downarrow} = \varepsilon$

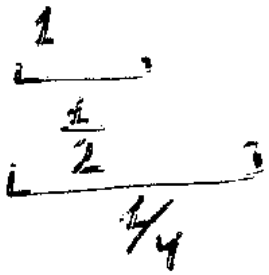
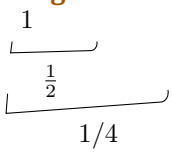


Diagram and typed description.



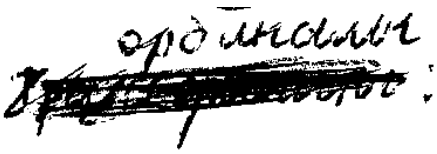
Original source details: GELFAND-P000789

$$\omega \cdot \varepsilon = 1$$
$$\omega + \varepsilon = \widehat{\uparrow} \uparrow \widehat{\downarrow}$$

Sometimes one can shorten it, but sometimes one cannot.

Ordinal arithmetic.

[crossed out: ...] [inserted above:] ordinals: numbers with arrows $\uparrow$ (without arrows $\downarrow$).



Undeciphered passage. The word below the insertion “ordinals” is heavily crossed out and illegible.

Original source details: GELFAND-P000789

The arithmetic is different from the ordinary one!

The ordinary one is not commutative!

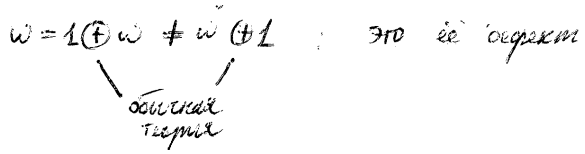


Diagram and typed description.
$\omega = 1 \oplus \omega \neq \omega \oplus 1$ This is its defect
ordinary theory

Original source details: GELFAND-P000789

Hessenburg devis. a natural def. of add. and mult. of transfin. numbers.



Uncertain reading. The last letters of the surname provisionally read “burg”.

Original source details: GELFAND-P000789



Uncertain reading. The abbreviation after the name provisionally reads «прид.» (devis.).

Original source details: GELFAND-P000789

It turned out that it coincid. with this one

$\forall$ number, except 0, has an inverse $\frac{1}{x}$, $x \neq 0$.

$\frac{y}{x}$ exists

$\forall x > 0$ there is $\sqrt{x}$ — by inf. ind.



Uncertain reading. The variable after $\forall$ is overwritten. The upper writing reads x.

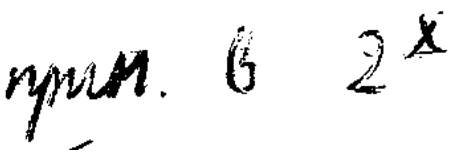
Original source details: GELFAND-P000789



Uncertain reading. The abbreviations after “by” provisionally read «инд. беск.» (inf. ind.).

Original source details: GELFAND-P000789

$\forall$ polynomial taking vals. of different signs at 2^{x} points becomes 0 somewhere between them; but this is inf. without contin. — has gaps, e.g., between finite and inf. numbers.



Uncertain reading. The abbreviation after “polynomial” provisionally reads «прин.» (“taking”). The numeral 2 is followed by a superscript Cyrillic x.

Original source details: GELFAND-P000789

[crossed out on the left: $\overset{?}{\uparrow}$] $\overset{\ell}{\uparrow}$ — “the largest” $\longrightarrow$ [inserted above:] already not a surreal number.



Uncertain reading. A loop at the top and several intersecting strokes are visible in the crossed-out sign.

Original source details: GELFAND-P000789



Uncertain reading. The looped sign above the arrow provisionally reads ℓ.

Original source details: GELFAND-P000789

$$\widehat{\uparrow\downarrow} = \frac{2}{3} \qquad \uparrow\downarrow\uparrow = \frac{3}{4}$$

$$\widehat{\uparrow}\overset{\ell}{\downarrow} \quad \text{— gap}$$



Uncertain reading. The looped sign above the down arrow provisionally reads ℓ.

Original source details: GELFAND-P000789

there are exp x; x^y; log

GELFAND-P000790

[source number: 6]

If x is pos. inf., then

$$2^x > x^n \qquad \text{for any } n.$$

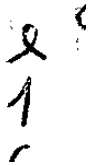
x — inf. small $\Rightarrow$

$$\exp x = \sum_{n \in \text{fin.}} \frac{x^n}{n!}$$

x — inf. small, n — real $\Rightarrow$ the series $\sum x^n a_n$ conv. $\forall a_n$

One possible appl. of the theory:

$$\int_x^{\overset{\ell}{\uparrow}} e^{-\xi^2}\, d\xi = \sum \frac{n!}{x^n}$$



Uncertain reading. The looped sign above the arrow at the integral’s upper limit provisionally reads ℓ.

Original source details: GELFAND-P000790

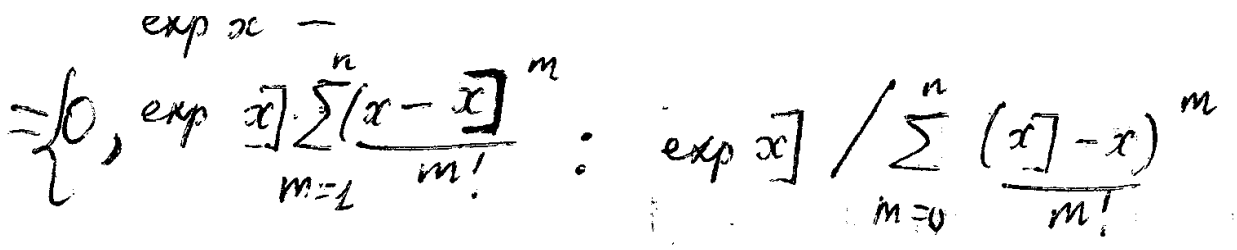
Every number = ratio of 2 integers $(= \frac{m}{n})$.

$$\widehat{\uparrow\downarrow} = \omega - 1$$

$$|x] - x| = 1 \qquad \text{— def. of an integer}$$

$$\exp x =$$

$$= \{0,\ \exp x] \cdot \sum_{m=1}^{n} \frac{(x-x]^m}{m!} :\ \exp x] \Big/ \sum_{m=0}^{n} \frac{(x]-x)^m}{m!}$$



Original source details: GELFAND-P000790

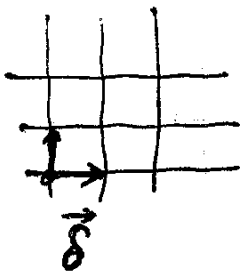
Uncertain reading. No closing parenthesis is visible after the square bracket in the first sum. The final closing brace is unwritten.

GELFAND-P000791

13.11.78. Gelfand Sem.

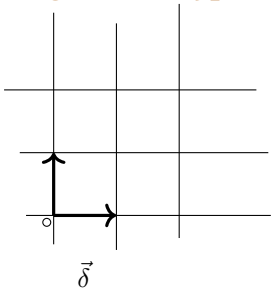
A. Polyakov 2-dimensional discr. system.

$$(1) \qquad S = \mathrm{Tr} \sum_{\vec{x},\vec{\delta}} g_{\vec{x}}^{-1} g_{\vec{x}+\vec{\delta}}$$



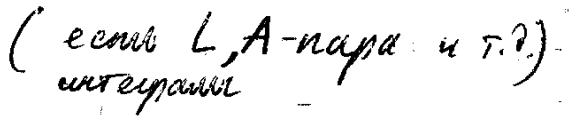
Original source details: GELFAND-P000791

Diagram and typed description.



$$(2) \qquad g_{\vec{x}} \in GL(n,\mathbb{R})$$
$$(3) \qquad \partial_\mu F_{\mu\nu}(x) + [A_\mu, F_{\mu\nu}] = 0$$
$$(4) \qquad \sum_\delta \left\{ \underbrace{g_{\vec{x}+\vec{\delta}} g_{\vec{x}}^{-1}}_{A_{\vec{x}\delta}} - g_{\vec{x}} g_{\vec{x}-\vec{\delta}}^{-1} \right\} = 0$$

The syst. is completely integr. and there is an inf. set of currents. (there is an L, A-pair and so on) [under “there is” is written: integrals]



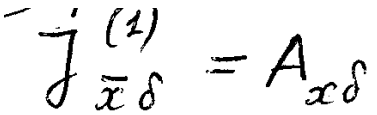
Uncertain reading. The final abbreviation provisionally reads «т. д.» (“so on”).

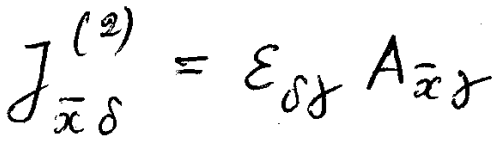
Original source details: GELFAND-P000791

$$(5) \qquad \sum_\delta \left(J_{\vec{x}\delta}^{(k)} - J_{\vec{x}-\vec{\delta},\delta}^{(k)} \right) = 0$$

Examples of currents:

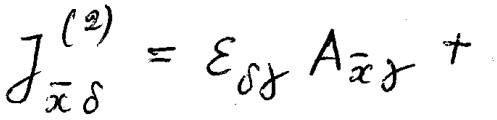
$$(6)_1 \qquad J_{\bar{x}\delta}^{(1)} = A_{x\delta}$$
$$(6)_2 \qquad J_{\bar{x}\delta}^{(2)} = \varepsilon_{\delta\gamma} A_{\bar{x}\gamma} +$$





Uncertain reading. Short bars are visible above x in the subscripts of $(6)_1$ and $(6)_2$. There is no bar over the x in $A_{x\delta}$.

Original source details: GELFAND-P000791



Undeciphered passage. The entry $(6)_2$ breaks off after the plus sign.

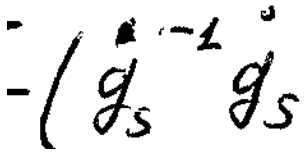
Original source details: GELFAND-P000791

For an ab. group and in the limit (cont.) in one var. — the Toda chain.

Here discr. time. and the nonab. case

Toda chain:

$$(7) \qquad \mathcal{L} = \mathrm{Tr} \sum_s \left(\dot{g}_s^{-1} \dot{g}_s + g_s^{-1} g_{s+1} \right)$$



Uncertain reading. A dot appears above the first g_s in (7), to the left of -1. The order of inversion and differentiation is unparenthesized.

Original source details: GELFAND-P000791

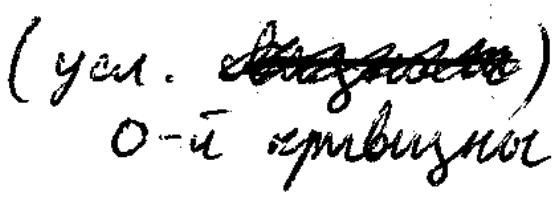
Let

$$(8) \qquad \begin{cases} A_s = g_s^{-1}\dot{g}_s \\ B_s = g_s^{-1}g_{s+1} \end{cases}$$

Zero-curvature conditions:

$$(9) \qquad \begin{cases} \dot{B}_s + A_sB_s - B_sA_{s+1} = 0 \\ \dot{A}_s = B_s - B_{s-1} \end{cases}$$

[to the right of the first line: (cond. of ~~connectedness?~~); inserted below: zero curvature]



Uncertain reading. The crossed-out word provisionally reads «связности» (“connectedness”).

Original source details: GELFAND-P000791

Find L_s, M_s, also satisf. the zero-curv. condition.

$$(10) \qquad \dot{L}_s + M_sL_s - L_sM_{s+1} = 0$$

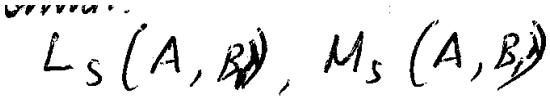


Uncertain reading. The sign after $\dot{L}_s$ is overwritten and provisionally reads as a plus.

Original source details: GELFAND-P000791

We want:

$$L_s(A, B, \lambda), \quad M_s(A, B, \lambda)$$



Uncertain reading. The signs by the closing parentheses provisionally read λ. Some strokes touch.

Original source details: GELFAND-P000791

such that $(10) \Rightarrow (9)$ and conversely.

Then

GELFAND-P000792

[source number: 17]

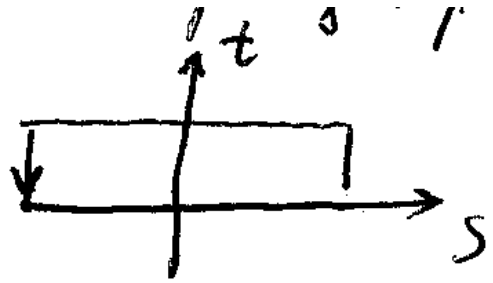
$$(11) \qquad \frac{d}{dt}\left\{\prod_{s=-\infty}^{\infty} L_s\right\} = 0$$



Uncertain reading. A short isolated dotted stroke under the product resembles i.

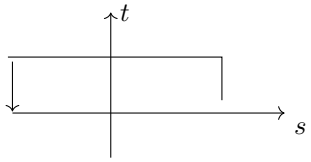
Original source details: GELFAND-P000792

generating fn. of all the integrals



Original source details: GELFAND-P000792

Diagram and typed description.



L, M — connections

$$I(0) \underset{\text{def}}{=} \prod L_s(0)$$

Since

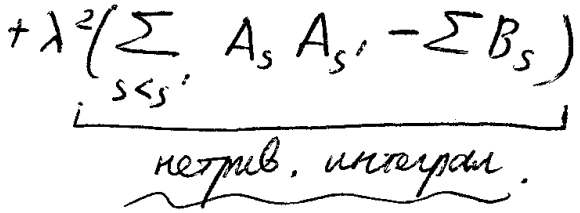
$$I(0)I^{-1}(\delta t) = 1,$$

then $I(\delta t) = I(0)$ and $I(t)$ — an integral of motion.
"Dumb way" of finding an L, A-pair: iterations in λ
answer.

$$(12) \qquad L_s = 1 + \lambda A_s - \lambda^2 B_s L_{s+1}^{-1}$$

Iter. in λ

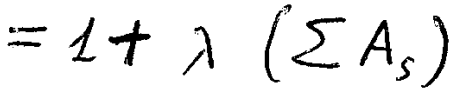
$$\prod_s (1 + \lambda A_s) = 1 + \lambda \left(\sum A_s \right)$$

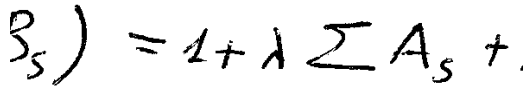


Original source details: GELFAND-P000792

Diagram and typed description.

$$(13) \quad \prod(1 + \lambda A_s - \lambda^2 B_s)$$
$$= 1 + \lambda \sum A_s + \lambda^2 \underbrace{\left(\sum_{s<s'} A_s A_{s'} - \sum B_s \right)}_{\text{nontriv. integral.}}$$

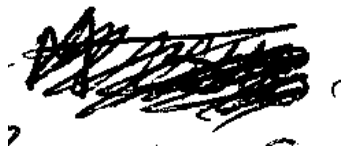




Original source details: GELFAND-P000792

Uncertain reading. Both expansions show equality signs. The subsequent powers are unwritten.

$$(14) \qquad \begin{cases} L_s = e^{-\partial} + A_s + B_s e^{\partial} \\ \dots \qquad M_s = B_s e^{\partial} \end{cases}$$



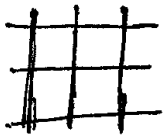
Original source details: GELFAND-P000792

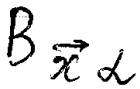
Undeciphered passage. The entry at the beginning of the second line of (14) is heavily crossed out. Its symbols are illegible.

Here e^{∂} — shift by 1 on the lattice

L_s, M_s — difference operators (Jacobi matr.)

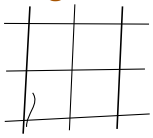
Gauge theory.





Original source details: GELFAND-P000792

Diagram and typed description.



$B_{\vec{x}\alpha}$

$\vec{x}$ — point, α — direction

($\alpha = 1, \dots, d$ —

— d — dimension)

2-dim. case — 2 dir.

(only positive ones are taken)

GELFAND-P000793

$$(15) \qquad F_{\bar{x}\alpha\beta} = B_{\bar{x},\alpha} B_{\bar{x}+\bar{\alpha},\beta} B^{-1}_{\bar{x}+\bar{\beta},\alpha} B^{-1}_{\bar{x},\beta}$$

Action:

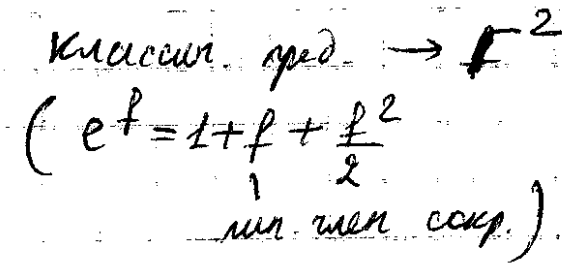
$$(16) \qquad S = \sum_{\bar{x},\alpha,\beta} \operatorname{Tr} F_{\vec{x}\alpha\beta}$$

[on the right:] Class. limit $\longrightarrow F^2$



Original source details: GELFAND-P000793

Uncertain reading. The sign before the raised 2 provisionally reads capital F.

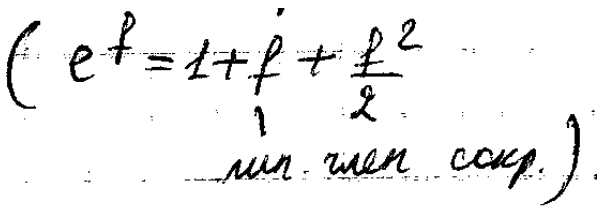


Original source details: GELFAND-P000793

Diagram and typed description.

($e^f = 1 + f + \frac{f^2}{2}$

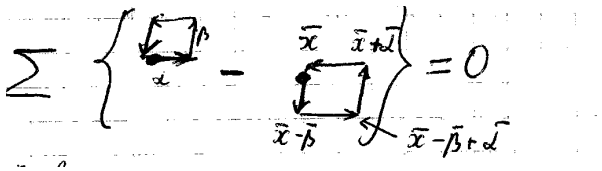
lin. term canc.).



Original source details: GELFAND-P000793

Uncertain reading. An equality sign is written in the parenthesis. The series ends after $f^2/2$.

Eqns. of motion:



Original source details: GELFAND-P000793

Diagram and typed description.
(17)

$$\sum \Big\{ \square_{\alpha}^{\beta} - \square_{\bar{x}}^{\bar{x}+\bar{\alpha}} \Big\} = 0$$

$\bar{x}-\bar{\beta}$ $\quad$ $\bar{x}-\bar{\beta}+\bar{\alpha}$

The dot marks the start

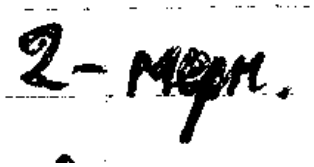
$\sum_\beta$, α — fixed (Cf.: $\nabla_\beta F_{\alpha\beta} = 0$)



Original source details: GELFAND-P000793

Uncertain reading. The overwritten sign provisionally reads ∇.

In the 2-dim. case there is no summation.



Original source details: GELFAND-P000793

Uncertain reading. The abbreviation for “dim.” is retraced. The earlier strokes are indistinct.

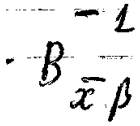
(17) means:

$$(17)_2 \quad \sum \{ B_{\bar{x},\alpha} B_{\bar{x}+\alpha,\beta} B^{-1}_{\bar{x}+\bar{\beta},\alpha} B^{-1}_{\bar{x},\beta} - B^{-1}_{\bar{x}\beta} B_{\bar{x}-\bar{\beta},\alpha} B_{\bar{x}-\bar{\beta}+\bar{\alpha},\beta} B^{-1}_{\bar{x}\alpha} \} = 0$$



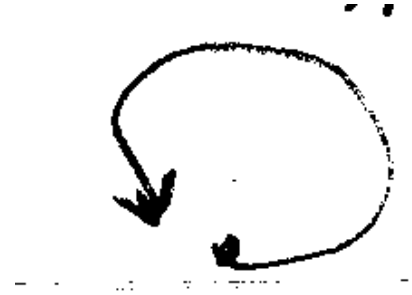
Original source details: GELFAND-P000793

Uncertain reading. The $\bar{x}$ in the last factor of the first product is retraced.



Original source details: GELFAND-P000793

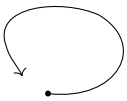
Uncertain reading. The index in the first inverse factor of the second product reads $\bar{x}\beta$.



Original source details: GELFAND-P000793

Diagram and typed description.

(18) $\Psi(c_{\bar{y}}) = \prod_{C_{\bar{y}}} B_{\bar{x}\beta}$





Original source details: GELFAND-P000793

Uncertain reading. The letter in the argument provisionally reads lowercase c. Its $\bar{y}$ subscript is retraced.



Original source details: GELFAND-P000793

Uncertain reading. The subscript below the circled product sign provisionally reads $C_{\bar{y}}$.

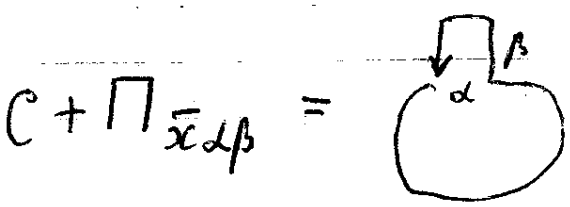


Original source details: GELFAND-P000793

Uncertain reading. The overwritten sign in the subscript of B provisionally reads $\bar{x}$.

Consider

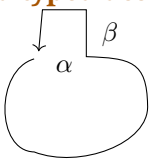
$$\Psi(c_y + \Pi_{\bar{x}\alpha\beta})$$



Original source details: GELFAND-P000793

Diagram and typed description.

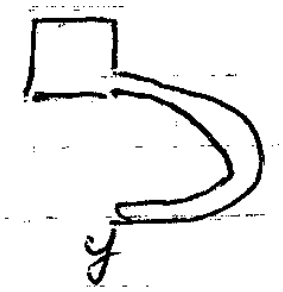
$C + \Pi_{\bar{x}\alpha\beta} =$



a small square has been attached

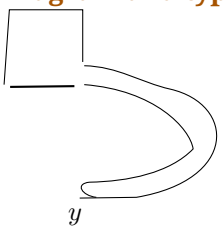
Now

$$\Psi(c+\Pi_{\bar{x}\alpha\beta})\Psi^{-1}(C)= \tag{19}$$



Original source details: GELFAND-P000793

Diagram and typed description.

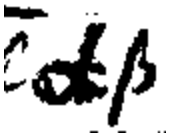


The gaps near the square and the thickened ends are shown. The directions at these points are unclear.



Original source details: GELFAND-P000793

Uncertain reading. The first sign resembles a looped lowercase c. The inverse function has argument C.



Original source details: GELFAND-P000793

Uncertain reading. The retraced subscript provisionally reads α.

dir. at pt. $\bar{x}, \alpha$, transported to point y

The eqns. are written in the form:

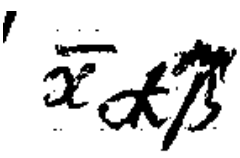
$$\sum_{\beta}\{\Psi(c+\Pi_{\bar{x}\alpha\beta})\Psi^{-1}(C) \tag{20}$$

$$-\Psi(C)\Psi^{-1}(C-\Pi_{\bar{x}\alpha\beta})\}=0$$



Original source details: GELFAND-P000793

Uncertain reading. The first contour is provisionally labelled with a lowercase c.



Original source details: GELFAND-P000793

Uncertain reading. The retraced pair of indices provisionally reads $\alpha\beta$.

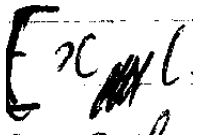
We get something like a connection (trivial) in the space of contours. (i.e. [inserted above the line:] matrix function of the contour)

(cf. (4))

Cont. case.

$\Psi[x_\mu(s)]$ — a functional of the curve (indep. of parametrization)

[on the right:] $x(s) = (x_\mu(s)); \quad \mu = 1, 2, 3, 4.$



Uncertain reading. The retraced subscript of x provisionally reads μ.

Original source details: GELFAND-P000793

Current

$$(21) \qquad \mathcal{F}_\mu(s, C) = \frac{\delta\Psi(x(s))}{\delta x_\mu(s)}\Psi^{-1}$$

GELFAND-P000794

[source number: 18]

Eqn.:

$$(22)_1 \qquad \frac{\delta\mathcal{F}_\mu(s, C)}{\delta x_\mu(s)} = 0$$

[on the right]

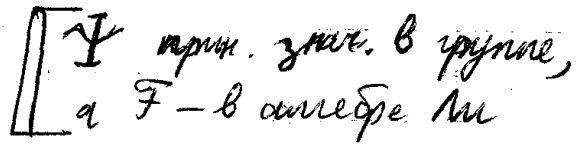


Diagram and typed description.

Ψ takes val. in the group,

and $\mathcal{F}$ — in the Lie algebra

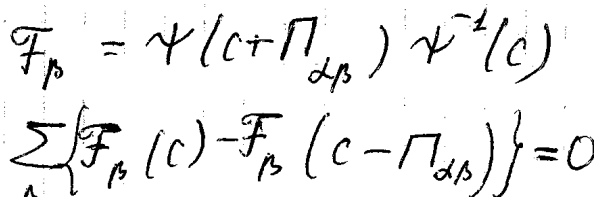
Original source details: GELFAND-P000794

Ψ is inv. with resp. to parametrization: this is written as

$$(23) \qquad \frac{dx_\mu}{ds}\frac{\delta\Psi}{\delta x_\mu(s)} = 0$$

$$\mathcal{F}_\beta = \Psi(c + \Pi_{\alpha\beta})\Psi^{-1}(c)$$

$$\sum_\beta \{\mathcal{F}_\beta(c) - \mathcal{F}_\beta(c - \Pi_{\alpha\beta})\} = 0$$



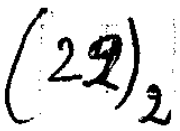
Uncertain reading. The letters in the arguments provisionally read lowercase c.

Original source details: GELFAND-P000794

the cont. limit gives (22).

$$\Psi = p\, e^{\oint A_\mu\, dx_\mu}$$

$$(22)_2 \qquad \frac{\delta \mathcal{F}_\mu(s)}{\delta x_\nu(s')} - \frac{\delta \mathcal{F}_\nu(s')}{\delta x_\mu(s)} + [\mathcal{F}_\mu(s), \mathcal{F}_\nu(s')] = 0$$

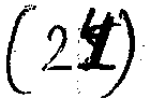


Uncertain reading. The second digit is overwritten. The upper writing provisionally reads $(22)_2$.

Original source details: GELFAND-P000794

Action

$$(24) \qquad S' = \int \left(\frac{\delta \Psi^{-1}}{\delta x_\mu(s)} \frac{\delta \Psi}{\delta x_\mu(s)} \right) (D x_\mu(s))$$



Uncertain reading. The second digit combines strokes resembling 4 and 1. Their order is unclear.

Original source details: GELFAND-P000794



Uncertain reading. The raised stroke to the right of S is typeset as a prime, S'.

Original source details: GELFAND-P000794

It is unclear how to define it

Carefully: 3 dimensions.

(in $4^{\underline{x}}$ dimensions it has yet to be checked in full).

The eqns. $(22)_1$ — $(22)_2$ are solved, with $\mathcal{F}$ obtained from formula (21) and Ψ satisf. cond. (23).

A Y.–M. field is a chiral field on the space of loops.

On the loops there is a fn. with val. in a finite-dimensional group. For it the eqns. of the principal chir. field (Ψ is such a field) are (22).

[inserted above the line on the left:] If Ψ comes from a connection ($\Leftrightarrow$ Ψ — homom. of loops into the group) $\Rightarrow$ one gets the Yang–Mills eqns..

Currents: ($\sim$ integrals of motion):

$$(25) \qquad \frac{\delta \mathcal{J}_\mu^{(k)}(s, C)}{\delta x_\mu(s)} = 0$$

$$\mathcal{J}_\mu^{(1)} = \mathcal{F}_\mu(s, C) \tag{26}$$

$$\mathcal{J}_\mu^{(2)} = \varepsilon_{\mu\nu\lambda} t_\nu(s) \mathcal{F}_\lambda(s, C) + [\mathcal{F}_\mu(s), \chi] \tag{27}$$

t — tang. vector to the curve

[$\mathcal{F}$] χ we shall now find



Uncertain reading. The crossed-out sign resembles $\mathcal{F}$.

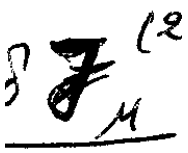
Original source details: GELFAND-P000795

We have:

$$\frac{\delta \mathcal{J}_\mu^{(2)}}{\delta x_\mu} = \varepsilon_{\mu\nu\lambda} t_\nu(s) \frac{\delta \mathcal{F}_\lambda(s, c)}{\delta x_\mu(s)} \tag{28}$$

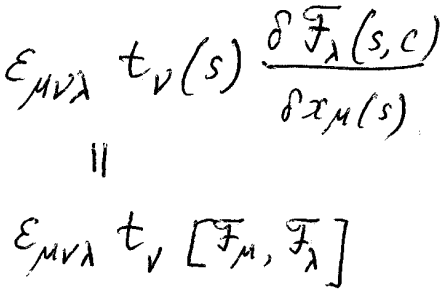
$$\| \varepsilon_{\mu\nu\lambda} t_\nu [\mathcal{F}_\mu, \mathcal{F}_\lambda]$$

$$+ \left[\mathcal{F}_\mu(s, c), \frac{\delta \chi}{\delta x_\mu(s)} \right]$$



Uncertain reading. The letter in the numerator on the left is overwritten. It may read $\mathcal{J}$ or $\mathcal{F}$.

Original source details: GELFAND-P000795



Uncertain reading. Two vertical strokes connect the first summand with the entry below it. No separate coefficient is visible before ε on the lower line.

Original source details: GELFAND-P000795

Let

$$\frac{\delta\chi}{\delta x_\mu} = -\varepsilon_{\mu\nu\lambda} t_\nu \mathcal{F}_\lambda \tag{29}$$

Then

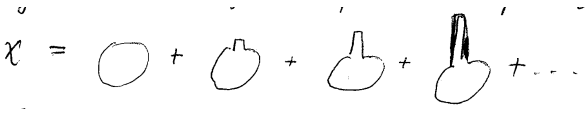
$$\mathcal{F}_\mu = \varepsilon_{\mu\nu\lambda} t_\nu \frac{\delta\chi}{\delta x_\lambda} \tag{30}$$

This is possible if the following holds $(22)_1$
We obtain a nonlocal conservation law. Expr. χ:



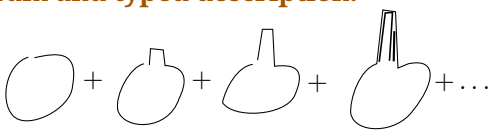
Original source details: GELFAND-P000795

Uncertain reading. The abbreviation before χ provisionally reads «Выраж-е.» (“expr.”). The dividing stroke is unclear.



Original source details: GELFAND-P000795

Diagram and typed description.

$\chi =$  $+\ldots$

The short gaps and thickened strokes on the last protrusion are shown. Arrowheads are indistinct.

(a sq. is added in one direction)
The nonlocality is 1-dimensional.
Contours: 1. S. Mandelstam, Phys. Rev., 1968.
2. Minlos, Sinai
[on the right, alongside item 1:] quant. Y.–M. in the spirit of Faddeev–Popov
γ-pencil (instead of the L, A-pair)



Original source details: GELFAND-P000795

Uncertain reading. The overwritten sign in the heading provisionally reads γ. λ is also possible.

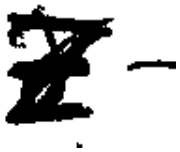
$$\frac{\delta\varphi}{\delta x_\mu(s)} + Z\,\varepsilon_{\mu\nu\lambda} t_\nu \frac{\delta\varphi}{\delta x_\lambda} = \mathcal{F}_\mu(s, C)\varphi(s, C) \tag{31}$$



Original source details: GELFAND-P000795

Uncertain reading. The overwritten parameter resembles a barred Z. Its letter case is unclear.

Z — arbitrary parameter



Uncertain reading. The letter in the explanation has a similar shape. Its case is unclear.

Original source details: GELFAND-P000795

The following has an analog. form:

Std. chiral field: (Pohlmayer, Zakharov, Mikhailov):

$$(32) \qquad A_\mu = g^{-1}\partial_\mu g(x);$$
$$[\partial_\mu + Z\,\varepsilon_{\mu\nu}\partial_\nu]\,\varphi = A_\mu\varphi, \qquad \mu = 1, 2, \ldots$$



Uncertain reading. The letter before ε provisionally reads Z. Its case is unclear.

Original source details: GELFAND-P000795

GELFAND-P000796

[source number: 19]

(31) — a system of eqns., Z — spectral param.



Uncertain reading. The letter in the explanation is retraced and resembles a barred Z. Its case is unclear.

Original source details: GELFAND-P000796

Compatibility of $(31) \Rightarrow$ eqns. (22).

$$Z = 0 \quad \Rightarrow \quad \varphi = \Psi$$



Uncertain reading. The letter before $= 0$ is retraced and resembles a barred Z. In the following formulas the letter is written smaller.

Original source details: GELFAND-P000796

Integrals — ?

In model (32) let us see how the integrals are obtained.

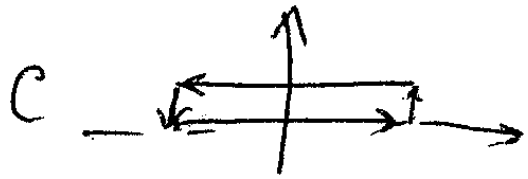
$$(32) \Leftrightarrow$$

$$(33) \qquad \partial_\mu \varphi = \frac{A_\mu + z\varepsilon_{\mu\nu}A_\nu}{1-z^2}\,\varphi$$

$$(34) \qquad P\exp \oint_C \frac{A_\mu + z\varepsilon_{\mu\nu}A_\nu}{1-z^2}\,dx^\mu = 1$$

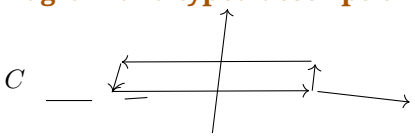
(ord. expon)

Spec. choice of contour:



Original source details: GELFAND-P000796

Diagram and typed description.



Additional strokes appear at the lower left corner. Their purpose is unclear.

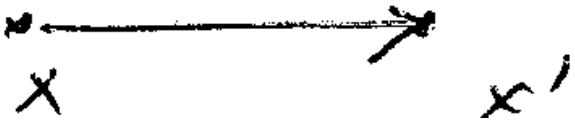
$$(35) \qquad \frac{d}{dt}\,P\exp \int_{-\infty}^{\infty} dx_1 \left\{ \frac{A_1 + zA_2}{1-z^2} \right\} = 0$$

This is the generating fn. of the integrals of motion.

The usual relations carry over to “contourology.”

There are Feynman diagrs. with surfaces in place of lines.

For free field theory



Original source details: GELFAND-P000796

Diagram and typed description.



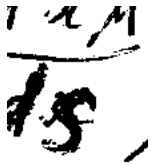
the Green fn. $G(x,x')$ satisfies the K.–G. eqn.

$$(36) \qquad (\partial_x^2 - m^2)G(x,x') = \delta(x-x')$$

$$(37) \qquad G(x,x') \underset{|x-x'|\to\infty}{\sim} e^{-m|x-x'|}$$

Here on contours

$$(38) \qquad \left\{ \frac{\delta^2}{\delta x_\mu^2(s)} - m^4 \left(\frac{dx_\mu}{ds} \right)^2 \right\} G(C) = 0$$



Original source details: GELFAND-P000796

Uncertain reading. The overwritten denominator provisionally reads ds.

$$(39) \qquad G(C) = \langle \Psi(C) \rangle$$

How does one find the contour Green fn.?

Math. problem: (38) — the "string" K.–G. eqn.
Boundary cond.: decay for large contours:

$$(40) \qquad G(C) \sim e^{-m^2 A(C)}$$

($A(C)$ — min. area spanning the contour C)

GELFAND-P000797

Let G depend only on the tensors

$$(41) \qquad \sigma_{\alpha\beta} = \oint_C x_\alpha \, dx_\beta$$

$$(42) \qquad \frac{\delta \sigma_{\alpha\beta}}{\delta x_\mu(s)} = \delta_{\alpha\mu} \frac{dx_\beta}{ds} - \delta_{\beta\mu} \frac{dx_\alpha}{ds}$$

$$t_\mu(s) = \frac{dx_\mu}{ds}$$

$$(43) \qquad \dddot{} \, t_\alpha t_\beta \left[\frac{\partial^2}{\partial \sigma_{\alpha\gamma} \partial \sigma_{\beta\gamma}} - m^4 \delta_{\alpha\beta} \right] G = 0$$



Uncertain reading. The crossed-out entry before $t_\alpha t_\beta$ resembles a summation sign with subscripts.

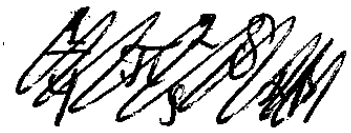
Original source details: GELFAND-P000797

Let

$$\pi_{\alpha\beta} = \frac{\partial}{\partial \sigma_{\alpha\beta}}$$

Then we obtain

$$(44) \qquad \pi_{\alpha\beta}\pi_{\beta\gamma} = \frac{1}{4}\pi^2 \delta \quad C\delta_{\alpha\beta};$$



Uncertain reading. The crossed-out signs after the equals sign resemble $\frac{1}{4}$, π^2 and δ. Their subscripts are unclear.

Original source details: GELFAND-P000797



Uncertain reading. The subscripts of the uncrossed δ on the right read $\alpha\beta$.

Original source details: GELFAND-P000797

This is possible only when

$$\pi_{\alpha\beta} = \pm {}^*\pi_{\alpha\beta} = \pm \varepsilon_{\alpha\beta\gamma\delta}\pi_{\gamma\delta} \tag{45}$$

Hence a fn. only of

$$U_\pm = \sqrt{\sigma_{\alpha\beta}^2 \pm \sigma_{\alpha\beta}\, {}^*\sigma_{\alpha\beta}} \tag{46}$$

Explicit solution:

$$G(U_+, U_-)$$

expressed via Bessel fns.

[between the lines:] Look at infinity:

Min. area

$$A_{\min}(C) = \sqrt{\sigma^2 \pm \sigma\, {}^*\sigma} \tag{47}$$

σ — projection onto coord. planes

$$\sigma_{\alpha\beta} = \oint x_\alpha \, dx_\beta$$

$$A = \int \sqrt{\dot{x}^2 x'^2 - (\dot{x}x')^2}\, dt\, ds$$

The Euler eqn. gives

$$A_{\min}(x(s))$$



Uncertain reading. The signs of the argument on the last line are retraced. They provisionally read $x(s)$.

Original source details: GELFAND-P000797

GELFAND-P000798

[source number: 20]

Ham.–Jacobi eqn.

$$\left(\frac{\delta A_{\min}}{\delta x_\mu(s)}\right)^2 = \left(\frac{dx_\mu}{ds}\right)^2$$

We find a solution. (47) — satisfies the Ham.–Jacobi eqn.

For contours that are not too [...] twisted, (47) — a solution.



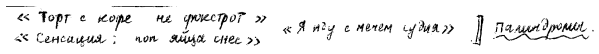
Undeciphered passage. A short entry between "too" and "twisted" is crossed out. The letters are indistinct.

Original source details: GELFAND-P000798

Sinai asserts that Courant has something.

GELFAND-P000799

20.11.78. Gelfand Sem.



Original source details: GELFAND-P000799

Diagram and typed description.
«Торт с кофе не фокстрот»
«Сенсация: поп яйца снес»
«Я иду с мечем судия»
Palindromes.
Cake with coffee is no foxtrot. Sensation: a priest laid eggs. I go with a sword, O judge.

Krichever: instantons without bundles.

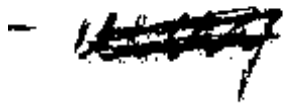
A. S. Schwarz.

1. Quantum [inserted above the line: field theory and] stat. phys. — the science of computing the integral (partition function)

$$Z = \int e^{-S(A)}\, dA \tag{1}$$

A ranges over the space of trajectories or histories: $A = A(x, t)$.

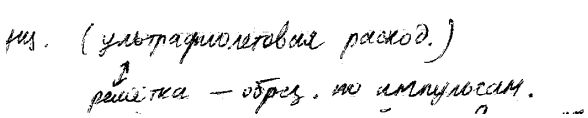
Meaning: we replace it by a finite-dimensional one (lattice approx.) and pass to the limit; a — [...] lattice spacing.



Original source details: GELFAND-P000799

Uncertain reading. A short word before “lattice spacing” is crossed out. It may read «шаг» (“step”).

QFT: the limit as $a \to 0$ does not exist.



Original source details: GELFAND-P000799

Diagram and typed description.
(ultraviolet divergences)
↙
lattice — momentum cutoff.

One way out is to take the limit for correl. fns., i.e. to compute

$$\frac{\int \varphi(A) e^{-S(A)}\, dA}{\int e^{-S(A)}\, dA} \tag{2}$$

(Green fns. or correlation fns.).

Important: the limit of the ratio already makes sense.

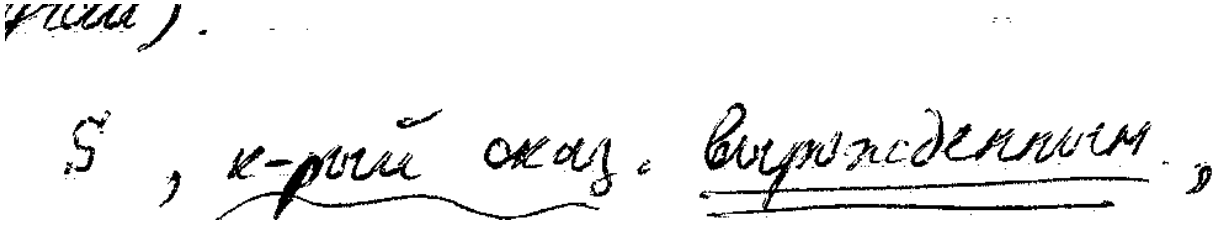
Also: when removing the cutoff one must do “renormalization”, i.e. the parameter λ_a must be varied as a function of a when passing to the limit.

There are 2 methods for computing cont. integrals:

perturbation theory and semiclassics

(in fact the 1st is a special case).

An important class of functionals S, which turns out to be degenerate, i.e. admitting an ∞-dimensional symmetry group:



Original source details: GELFAND-P000799

Uncertain reading. The abbreviation before «вырожденным» (“degenerate”) reads «оказ.». A faint dot is visible before the comma after the word.

$$g \in G \quad \rightsquigarrow \quad A \mapsto A^g$$
$$S(A^g) = S(A) \tag{3}$$

Physical examples: gauge theories.

Pure Y.–M. theory; $A = (A_\mu(x))$, $A_\mu \in$ Lie alg. of a finite-dimensional Lie group.

$$S(A) = \int \langle F_{\mu\nu}, F^{\mu\nu} \rangle\, dV, \tag{4}$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \tag{5}$$



Original source details: GELFAND-P000799

Uncertain reading. The subscript of the second ∂ is heavily retraced. It provisionally reads ν.

GELFAND-P000800

[source number: 26]

Symmetry group:
if $g(x)$ — a fn. with values in the Lie group, then it generates a transformation of the fields

$$A_\mu \mapsto g A_\mu g^{-1} + g\,\partial_\mu(g^{-1}) \tag{6}$$

[crossed out above the next line: ...]



Original source details: GELFAND-P000800

Undeciphered passage. A short entry above «Способ» (“Method”) is obscured by heavy hatching.

Method for dealing with degeneracy. In the case of the electromagn. field.
“The potentials have no meaning”, let us impose a gauge condition

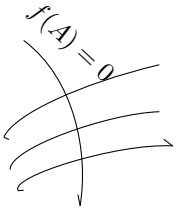
$$\partial_\mu A_\mu = 0 \tag{7}$$

Difficulties: it is hard to construct a perturbation theory.
2. Semiclassics $\sim$ stationary phase or Laplace method.
a) A problem solved long ago: how to write the cont. integral (Faddeev and Popov)
[below the line:] — at the physicist’s level of rigor.
b) Math. theory — when S is quadratic.
c) Application to computing the instanton contribution.
d) Ray–Singer invariants.
a) Quantization of degen. Lagrangians.
Difficulty: the integral diverges, since the volume of the orbits is infinite.
moral: choose a gauge condition — a transversal to the orbits — and integrate over it.



Original source details: GELFAND-P000800

Diagram and typed description.



Angled strokes are visible at the right end of the lowest curve and the bottom of the transverse curve. The redraw shows the visible sides of those strokes.

but: one must insert the factor $\delta(f(A))$.

In addition, one must insert the Jacobian $\Delta_f(A)$

$$\Delta_f(A)\delta(f(A)) \tag{8}$$

The F.–P. method is to compute $\Delta_f(A)$. (he does not go into this)

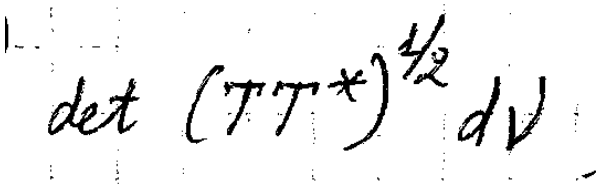
$f(A) = 0$ — eqn. of the transversal.

For finite-fold integrals, here — a theorem.

[added on the left:] Another method.

Let us consider the integral again (in the finite-fold case)

$$\int_M e^{-S(A)}\, dA = \int_{M/G} e^{-S(A)} \det(TT^*)^{1/2}\, dV, \tag{9}$$



Uncertain reading. Equation (9) reads $\det(TT^*)^{1/2}$. No prime is visible on det.

Original source details: GELFAND-P000800

where dV — the measure on the set of orbits, obtained from the std. Riem. metric on it.

T — the operator obtained as follows: take (assuming the action free):

$$T: \underset{\text{Lie alg.}}{\mathfrak{g}} \xrightarrow{T_x} T_x(M)$$

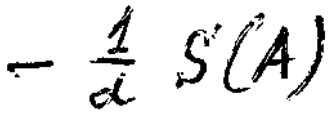
[on the right:] map obtained from the group action.

Assume that G acts by isometries on M.

GELFAND-P000801

In physics the problem arises of computing by Laplace's method, i.e. of finding the asymptotics of

$$\int e^{-\frac{1}{\alpha}S'(A)}\, dA \qquad \text{as } \alpha \to 0 \tag{10}$$



Uncertain reading. A separate short stroke follows S in the exponent of (10). It is provisionally read as S'.

Original source details: GELFAND-P000801

One needs to know the stat. points of $S'(A)$, called instantons.



Uncertain reading. The short stroke before the opening parenthesis is provisionally read as a prime on S.

Original source details: GELFAND-P000801

The math. problem is as follows. There is a set of stat. points (even several manifolds), and even ∞-dimensional ones. There is also strong degeneracy.

Examples:

1) Y.–M. field. $G = SU(2)$

$\forall q$ there is a family of $8q - 3$ instantons apart from the ∞-dimensional symmetry.

$$2) \qquad S(\vec{n}) = \frac{1}{f_0} \int (\partial_\mu \vec{n})^2 \dots d^2x, \quad \mu = 0, 1; \quad x = (x_0, x_1) \tag{11}$$



Undeciphered passage. An entry between $(\partial_\mu \vec{n})^2$ and d^2x is heavily crossed out. A loop is visible at the left. The entry itself is illegible.

Original source details: GELFAND-P000801

One has to compute Gaussian integrals, and there arises det.

A degen. Lagrangian gives one more det — from Faddeev–Popov — it is $\det(TT^*)^{1/2}$.

The determinants are ∞-dimensional. They require analytic continuation or renormalization.

And they turn out to depend on the choice of metric.

But one can trace the change in det under variation of the metric.

In (11): $\vec{n} = (n_1, n_2, n_3) \in S^2, \quad |\vec{n}| = 1.$

The extremals in (11) were found by Belavin and Polyakov.

Let us pass to stereogr. variables on the plane — to the sphere.

We obtain $w(z)$.

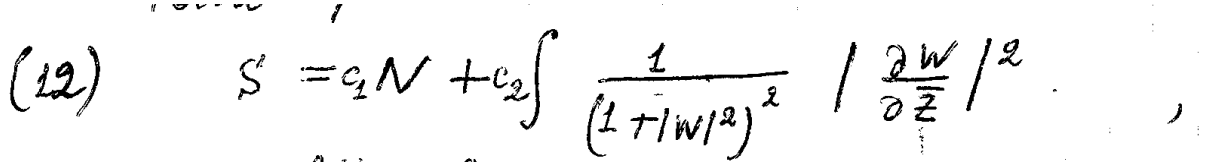
Answer: extremals $\leftrightarrow w(z)$ — rational fns. of z!

Explanation: there is a top. number N — the degree of the map

($\sim$ integral of the Jacobian).

Then a simple calculation shows

$$S' = c_1 N + c_2 \int \frac{1}{(1 + |w|^2)^2} \left| \frac{\partial w}{\partial \bar{z}} \right|^2, \tag{12}$$



Original source details: GELFAND-P000801

Uncertain reading. A short stroke is visible beside S on the left of (12). It is provisionally read as S'. No differential is visible after the squared modulus.

from which everything follows.

One must compute det.

There is no gauge group here and there is one det. The asympt. of (10) is given by the

integral

$$\int \det{}^{-1/2}(\dots)\, d\mu_0, \tag{13}$$

where μ_0 — some measure on instantons.

~~It turns out to be convenient~~

The operator is constructed from $\forall$ rational fn. as the [inserted above the line: op. of the quadr. form corresp. to] the principal quadr. part of the f'nal

$$z = x_0 + ix_1$$

$$z \xrightarrow{w} \mathrm{CP}(1) \tag{14}$$

GELFAND-P000802

[source number: 27]

$T\mathrm{CP}(1)$ induces a bundle on the z-plane (by this map).

[below $T\mathrm{CP}(1)$:] (holomorphic)

[on the right:] w — rat'l.

$$\Delta = \bar\partial^* \bar\partial, \qquad \bar\partial = \frac{\partial}{\partial \bar z} : \Lambda^{0,0} \to \Lambda^{0,1} \tag{15}$$

[under the right-hand part:] on forms with values in the bundle.

The metric also depends on w.

(induced by the std. metric of CP^1)

w actually enters only in the $*$.

One has to compute det. This is done using the general theory.

[crossed-out block: ...]



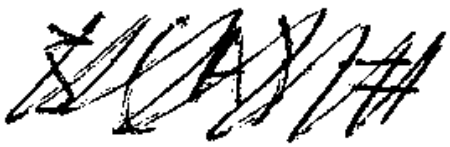
Undeciphered passage. Dense horizontal and diagonal strokes fill the outlined block. The underlying entry is illegible.

Original source details: GELFAND-P000802

det is regularized, simultaneously renormalizing f_0 (coupling const.) in (11).

One way to regularize is to consider

[crossed out: ~~$\zeta(A) =$~~]



Uncertain reading. The crossed-out entry is provisionally read as $\zeta(A) =$. Cancellation strokes intersect its ending.

Original source details: GELFAND-P000802

$$\zeta(s) = \sum_{\lambda_i \neq 0} \lambda_i^{-s} \tag{16}$$

$s = 0$ — point of analyticity.

$$\ln\det \underset{\text{def}}{=} -\zeta'(0). \tag{17}$$

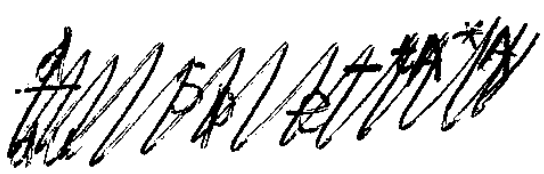
The op. $\bar{\partial}$ is “one and the same” for all instantons with the same top. number N.

General statement: — when the metric changes, one can trace the change in

$$\det(A^*A)$$

idea:

[$\frac{d}{du}\operatorname{Sp} e^{-tA^*A}$]



Uncertain reading. The crossed-out entry is provisionally read as $\frac{d}{du}\operatorname{Sp} e^{-tA^*A}$. Diagonal strokes intersect the exponent and denominator.

Original source details: GELFAND-P000802

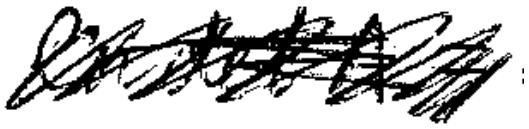
$$\frac{d}{du}A^*A = BA^*A + A^*CA \tag{18}$$

Here B arises from the formula giving the variation of the scalar product:

$$(u, v) = \langle u, v\rangle + \langle Bu, v\rangle. \tag{19}$$

The variation of $\det A^*A$ will be expressed in terms of $\operatorname{Sp} e^{-tA^*A}$ using the Frullani integral.

[crossed out above the next formula: ~~$\ln\det A =$~~]



Uncertain reading. The crossed-out line is provisionally read as $\ln\det A =$. The final strokes are obscured.

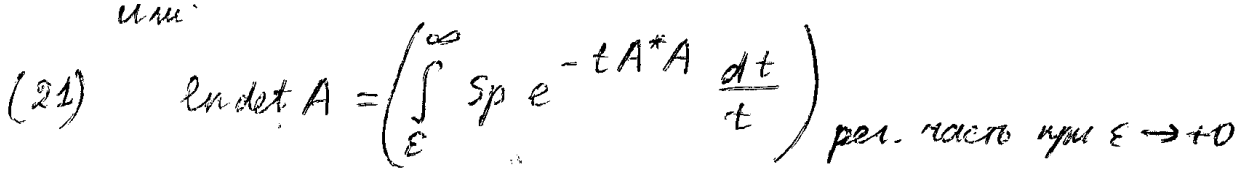
Original source details: GELFAND-P000802

$$\int_0^\infty \frac{e^{-ta} - e^{-tb}}{t}\, dt = \ln\frac{b}{a} \tag{20}$$

(Frullani integral).

or:

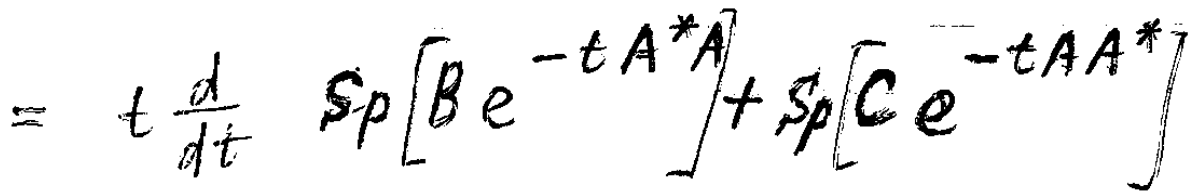
$$\ln\det A = \left(\int_\varepsilon^\infty \operatorname{Sp} e^{-tA^*A}\frac{dt}{t}\right)_{\text{reg. part as } \varepsilon\to+0} \tag{21}$$



Original source details: GELFAND-P000802

Uncertain reading. The left side of (21) reads $\ln\det A$. No minus sign or numerical factor is visible before the integral.

$$\begin{aligned}\frac{d}{du}\operatorname{Sp} e^{-tA^*A} &= -t\operatorname{Sp}\big[(BA^*A + A^*CA)e^{-tA^*A}\big] = \\ &= t\frac{d}{dt}\operatorname{Sp}\big[Be^{-tA^*A}\big] + \operatorname{Sp}\big[Ce^{-tAA^*}\big]\end{aligned} \tag{22}$$



Original source details: GELFAND-P000803

Uncertain reading. In the second line of (22) the bracket after the trace with B closes before the plus sign. The trace with C is written separately, with no visible $t\,d/dt$.

t — cancels with $\frac{1}{t}$ in (21)

and integrals come out, the corresp. coeffs. of the Seeley asympt.

$$\ln\det A = \int_\varepsilon^\infty \operatorname{Sp} e^{-tA^*A}\frac{dt}{t} + \sum a_i\varepsilon^i + a_0\ln\varepsilon.$$

What matters is a_0.

$t\to\infty \quad\Rightarrow\quad$ zero modes.

this part is left unevaluated.

One has to compute the integral with respect to $d\mu_0$.

$d\mu_0$ has to be computed too! — it is an n-dimensional det

One has to compute the $(4q+2)$-dimensional det of the metric tensor.

The metric on rat'l fns. comes from the std. metric on all fns. (in their tangent space) $=\sum$ squares of variations.

In fact the **2** unevaluated parts (from the modes and from $d\mu_0$) cancel!!

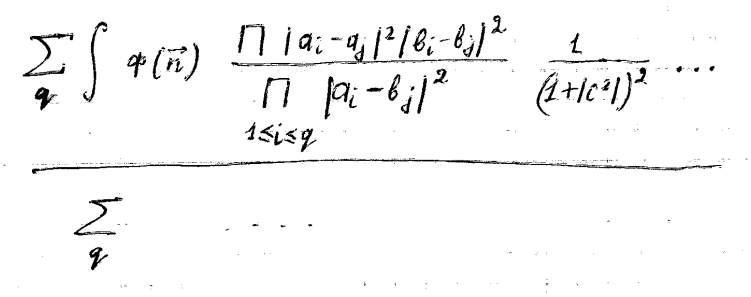
One still has to compute

$$\int \Phi(\vec n)\,\overset{-1/2}{\det}(\dots)\,d\mu_0.$$

Specify a rat'l fn. by its zeros and poles.

$$\begin{array}{l|l} a_1\dots a_n & \text{— zeros} \\ b_1\dots b_n & \text{— poles} \end{array} \quad c \text{ — asympt. at } \infty.$$

$$\frac{\sum_q \int \varphi(\vec n)\dfrac{\prod|a_i-a_j|^2|b_i-b_j|^2}{\prod_{1\le i\le q}|a_i-b_j|^2}\dfrac{1}{(1+|c^2|)^2}\cdots}{\sum_q \cdots}$$



Original source details: GELFAND-P000803

Uncertain reading. The upper product has no written limits. The lower one has $1\le i\le q$ beneath it, while both i and j occur in the factor. The exponent in $|c^2|$ is inside the modulus bars.

$$\varphi(\vec{n}) = n_\alpha(z')n_\beta(z'')$$

The denominator is the partition function of a Coulomb gas in the plane.

At poles — $+$ charges

at zeros — $-$ charges.

[on the right:]

$$\left(\prod_{\dots} = \dots e^{\sum \ln \dots}\right)$$

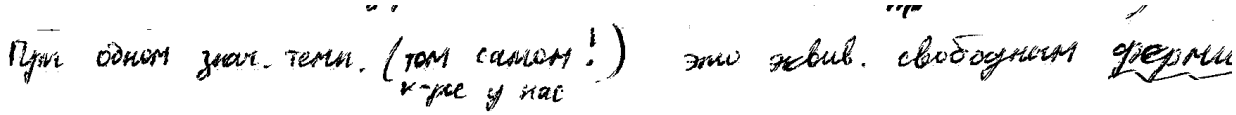


Undeciphered passage. The sign before e is heavily crossed out. Angled strokes remain visible and the sign is illegible.

Original source details: GELFAND-P000803

At one value of the temp. (that very one!) this is equiv. to free fermions.

[inserted below: the one we have]



Original source details: GELFAND-P000803

Uncertain reading. The abbreviation in the note below the parenthesis is provisionally read as «к-рое» (“which”, referring to the value).

GELFAND-P000804

27.11.78. Gelfand Sem.

Lev Nikolaevich Lipatov

on the asympt. of diagrams.

Zakharov.

For integrable systems “in general position” the var. principle will hold.

(all the known ones are out of general position)

$$\begin{cases} \Psi_\xi = U\Psi \\ \Psi_\eta = V\Psi \end{cases} \tag{1}$$

$$\left.\begin{aligned} U &= U(\xi,\eta,\lambda) \\ V &= V(\xi,\eta,\lambda) \end{aligned}\right\} \quad \text{rat'l fns. in } \lambda$$

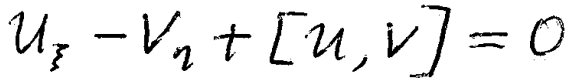
$$\begin{cases} U = U_0 + \sum\limits_{n=1}^{N_1} \dfrac{U_n}{\lambda - \lambda_n}, \\ V = V_0 + \sum\limits_{n=1}^{N_2} \dfrac{V_n}{\lambda - \mu_n} \end{cases} \tag{2}$$

$$\lambda_i \neq \lambda_j; \qquad \mu_i \neq \mu_j; \qquad \lambda_n \neq \mu_m$$

Ψ, U, V — matrices

Int. system:

$$U_\xi - V_\eta + [U,V] = 0 \tag{3}$$



Uncertain reading. In (3) the subscript on U is ξ, and the one on V is η.

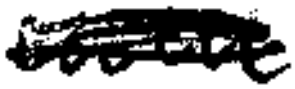
Original source details: GELFAND-P000804

Lax pairs: U has a simple pole at the 1st point, V has a pole of any order at that same point

GELFAND-P000805

[source number: 37]

[crossed out at the start of the line: ...]



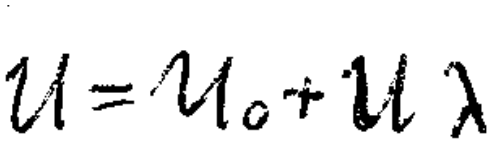
Undeciphered passage. Several horizontal strokes cover a short word. Its letters are indistinct.

Original source details: GELFAND-P000805

Then U, V can be made into polynomials:

$$U = U_0 + U\lambda$$

$$V = P_n(\lambda)$$



Uncertain reading. No subscript is visible on the U before λ.

Original source details: GELFAND-P000805

In more detail, (3) has the form:

$$\begin{cases} U_{0\eta} - V_{0\xi} + [U_0, V_0] = 0 \\ U_{n\eta} = [U_n, A_n] \\ V_{n\xi} = [V_n, B_n] \end{cases} \tag{4}$$

Here

$$A_n = V\big|_{\lambda=\lambda_n}, \qquad B_n = U\big|_{\lambda=\mu_n}. \tag{5}$$

This is a syst. of $N_1 + N_2 + 1$ eqns. for $N_1 + N_2 + 2$ quantities.

Let us integrate partially. Let $U_n^{(0)}$ and $V_n^{(0)}$ — Jordan forms, then

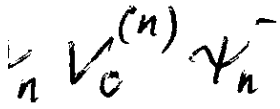
$$U_n^{(0)} = U_n^{(0)}(\xi), \qquad V_n^{(0)} = V_n^{(0)}(\eta) \qquad \text{(by } (4)_{2,3}\text{)} \tag{6}$$

$$\begin{cases} U_n = \varphi_n U_n^{(0)} \varphi_n^{-1}, \\ V_n = \psi_n V_0^{(n)} \psi_n^{-1} \end{cases} \tag{7}$$



Uncertain reading. The heavily retraced number provisionally reads (7). Earlier strokes within the digit are indistinct.

Original source details: GELFAND-P000805



Uncertain reading. The middle V on the second line of (7) has subscript 0 and superscript (n).

Original source details: GELFAND-P000805

We take the matrices in (6) as given.
The syst. will be integrated if we find φ_n and ψ_n in (7).
φ_n, ψ_n are determined nonuniquely. One representative is needed.
Set:

$$\nabla_\eta = \frac{\partial}{\partial \eta} - V_0, \qquad \nabla_\xi = \frac{\partial}{\partial \xi} - U_0. \tag{8}$$

Then the eqns. take the form

$$\begin{cases} \left(\nabla_\eta - \sum\limits_{m=1}^{N_2} \dfrac{V_m}{\lambda_n - \mu_m} \right) \varphi_n = 0 \\ \left(\nabla_\xi - \sum\limits_{m=1}^{N_1} \dfrac{U_m}{\mu_n - \lambda_m} \right) \psi_n = 0 \end{cases} \tag{9}$$

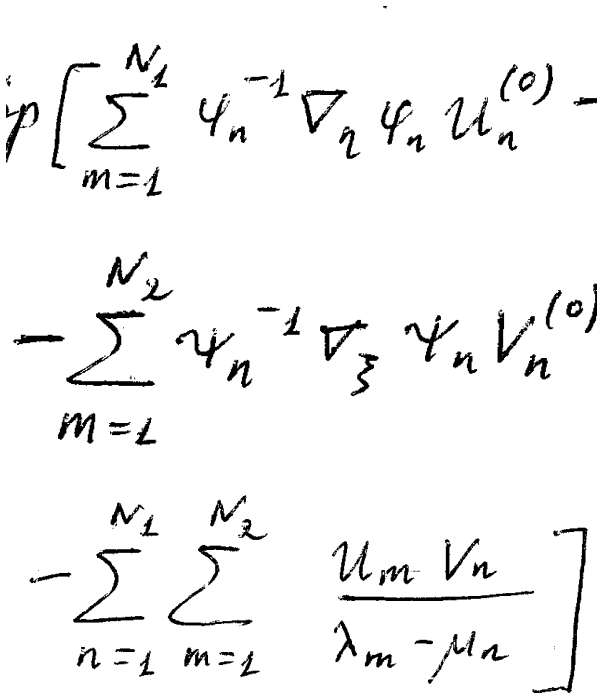
[on the right:] (it is understood that V_m, U_m are expr. by (7))
Here we have class. spinor models (for physicists). Action:

$$S' = \int d\xi \, d\eta \ \mathrm{Sp} \Big[\sum_{m=1}^{N_1} \varphi_n^{-1} \nabla_\eta \varphi_n U_n^{(0)} - \sum_{m=1}^{N_2} \psi_n^{-1} \nabla_\xi \psi_n V_n^{(0)} = - \sum_{n=1}^{N_1} \sum_{m=1}^{N_2} \frac{U_m V_n}{\lambda_m - \mu_n} \Big] \tag{10}$$



Uncertain reading. A raised stroke beside S in (10) is provisionally read as a prime, S'.

Original source details: GELFAND-P000805



Uncertain reading. The first two sums start at $m = 1$, while the factors use index n. The double sum has $n = 1$ beneath N_1 and $m = 1$ beneath N_2. The fraction reads $U_m V_n / (\lambda_m - \mu_n)$.

Original source details: GELFAND-P000805



Uncertain reading. Two strokes at the end of the first line are provisionally read as =. The next line begins with a minus sign.

Original source details: GELFAND-P000805

GELFAND-P000806

Hence series of conservation laws (related to the number of poles)

$N_1 + N_2$ series.

Simplest syst. — sine–Gord. — poles at 0 and ∞.

Conjecture: there are many Ham. structures.

(as for KdV — an infinite number of Ham. structures)

Lipatov

ZhETF 72, 411, 1977.

$$\alpha = \frac{e^2}{4\pi\hbar c} = \frac{1}{137}$$

$$\mathcal{L} = \int d^3x\, dt \left[\frac{[\partial_\mu \varphi]^2}{2} - \frac{m_0^2 \varphi^2}{2} - \lambda_0 \frac{\varphi^4}{4!} \right] \tag{1}$$

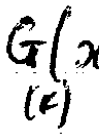
Problem: [...] computing Green fns. (in the Euclidean region):



Undeciphered passage. A heavily inked sign follows «Задача» (“Problem”). Its form is illegible.

Original source details: GELFAND-P000806

$$G_{(c)}(x_1, \ldots, x_n) = Z_0^{-1} \int \prod_x d\varphi(x)\, e^{-H(\varphi)} \prod_{i=1}^{n} \varphi(x_i) \tag{2}$$



Uncertain reading. The parenthesized subscript is provisionally read as (c). It has an angled stroke open at the right.

Original source details: GELFAND-P000806

$$H(\varphi) = \int d^4x \left[\frac{(\partial_\sigma \varphi)^2}{2} + \frac{\lambda_0 \varphi^4}{4!} \right] + \frac{m_0^2 \varphi^2}{2} \Bigg]$$



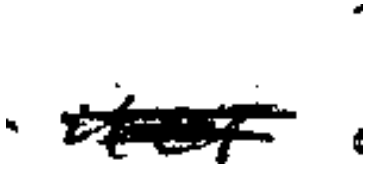
Uncertain reading. The closing square bracket after the fraction with λ_0 is crossed out. A further bracket stands after the final summand.

Original source details: GELFAND-P000806

(= $\mathcal{L}$ in the Euclidean region)

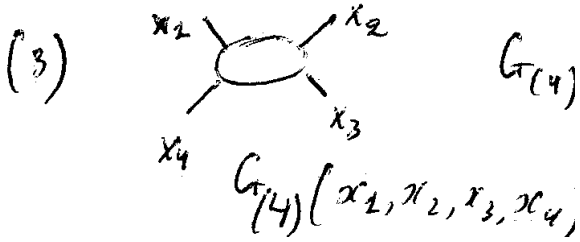
Integral (2) diverges. But if we make λ_0, m_0 dep. on the cutoff parameters, the answer will be finite (renormalizations!)

m_c^2 — pole of the Green fn. (m_c — [...] phys. mass).



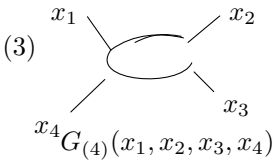
Original source details: GELFAND-P000806

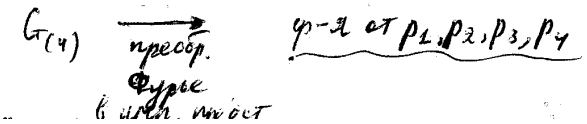
Undeciphered passage. A short entry before "phys. mass" is crossed by several strokes. Its letters are indistinct.



Original source details: GELFAND-P000806

Diagram and typed description.





Original source details: GELFAND-P000806

Diagram and typed description.

$G_{(4)} \xrightarrow[\text{transf.}]{\text{Fourier}}$ fn. of p_1, p_2, p_3, p_4
in mom. space

(∼ scattering of 4 particles)

Pole = [...] m_c^2



Original source details: GELFAND-P000806

Uncertain reading. The whole crossed-out expression before m_c^2 is illegible. Its initial strokes may belong to m_0.

fn. (3) depends only on p_j^2 and $p_i \cdot p_j$ (by Lorentz invariance)

and contains $\delta(p_1 + p_2 - p_3 - p_4)$ (by momentum conservation)

Vertex Green fn.:

$$\Gamma = \frac{G(p_1, p_2, p_3, p_4)}{G(p_1)G(p_2)\dots G(p_4)} \tag{4}$$

(after factoring out the δ-fn.)

By perturbation theory (with $\lambda_0 = 0$) we obtain

$$G^0(p) = \frac{1}{p^2 + m^2}. \tag{5}$$

GELFAND-P000807

[source number: 38]

[crossed out: ...]

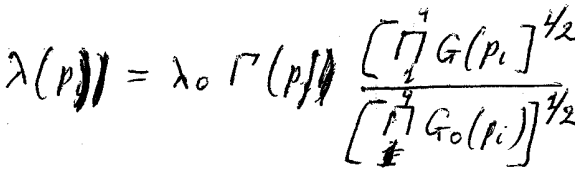
$$\frac{[G(p_1)\dots G(p_4)]^{1/2}}{[G^0(p_1)\quad G^0(p_4)]^{1/2}}$$



Undeciphered passage. A heavily crossed-out entry before the upper fraction may read $\Gamma(p)$. The strokes inside the parentheses are indistinct.

Original source details: GELFAND-P000807

$$\lambda(p_i)) = \lambda_0\Gamma(p_i))\frac{[\prod_1^4 G(p_i]^{1/2}}{[\prod_1^4 G_0(p_i)]^{1/2}} \quad \text{— charge (invariant)}$$



Uncertain reading. The arguments of λ and Γ are provisionally read as p_i with two closing-parenthesis strokes. In the numerator a square bracket follows p_i with no separate closing parenthesis visible.

Original source details: GELFAND-P000807

$$p = (p_1, p_2, p_3, p_4)$$

λ_0 is chosen so that the phys. charge is finite (and all the Green fns. too)

The paradox of Landau and Pomeranchuk: if we set perturb. theory aside and keep the leading ln, then we get

$$\lambda_\mu = \frac{\lambda_0}{1 + \frac{3}{2}\frac{\lambda_0}{16\pi^2} \dots \ln\frac{\Lambda}{\mu^2}}.$$

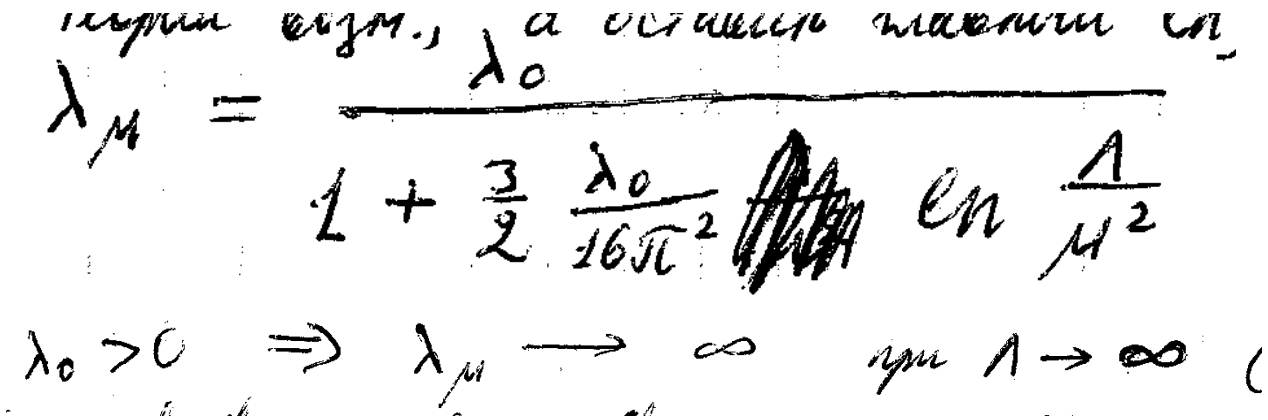


Undeciphered passage. The entry before ln is heavily crossed out. Its symbols are indistinct.

Original source details: GELFAND-P000807

[to the right of the formula:] μ — defining point: $p_i p_j = \mu^2$, $p_i^2 = (\dots)\mu^2$.
$(\cdot)$ from $(\sum p_i)^2 = 0$.

$$\lambda_0 > 0 \quad \Rightarrow \quad \lambda_\mu \longrightarrow \infty \quad \text{as } \Lambda \longrightarrow \infty \quad \text{(i.e. the cutoff is removed).}$$



Original source details: GELFAND-P000807

Uncertain reading. The fraction has a plus sign after 1. Below, $\lambda_\mu \to \infty$ is written for $\Lambda \to \infty$.

The same in quant. electrodyn. [~~...~~] (for no bare λ_0 does one obtain $\frac{1}{137}$)



Undeciphered passage. Diagonal strokes cross out an entry between "electrodyn." and the opening parenthesis. It is illegible.

Original source details: GELFAND-P000807

One needs the asymptotics of the coeffs. of the perturbation series arising for

$$\psi = \frac{d\lambda_\mu}{d \ln \frac{\Lambda}{\mu^2}} = a_2 \lambda_\mu^2 + a_3 \lambda_\mu^3 + \dots$$

the asympt. in k as $k \to +\infty$ exists
The first 4 have been found explicitly.
Idea: Laplace's method for the continuum integral
Sobolev inequality:

$$\left[\int (\partial_\sigma \varphi)^2 \, d^4x \right]^2 \geq \frac{32\pi^2}{3} \int \varphi^4(x) \, d^4x$$

equality is attained at

$$\varphi = \frac{c_1}{(x - x_0)^2 + \lambda^2}$$

[Zakharov's insertion $\Delta\varphi + \varphi^3 = 0$ — obtained from the Y.–M. eqns.]

GELFAND-P000808

4.12.78 Gelfand Sem.

A. Guichardet

Extensions of reps. of top. groups.

1. Exts. of modules over discr. groups.

E, F — 2 G modules.



Uncertain reading. The heavily retraced digit between the dash and G provisionally reads 2.

Original source details: GELFAND-P000808

An ext. of E by F — is a third G-module H, such that $H \supset F$ and $H/F \cong E$ or an exact seq.

$$0 \longrightarrow F \longrightarrow H \longrightarrow E \longrightarrow 0 \tag{1}$$

2 exts. are equiv. if $\exists$ an isom. such that the diag. commutes.

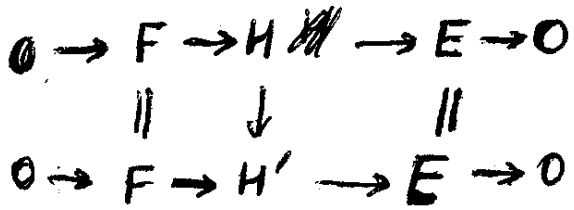


Diagram and typed description.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & F & \longrightarrow & H & \cdots\!\!\longrightarrow & E & \longrightarrow & 0 \\ & & \| & & \downarrow & & \| & & \\ 0 & \longrightarrow & F & \longrightarrow & H' & \longrightarrow & E & \longrightarrow & 0 \end{array}$$

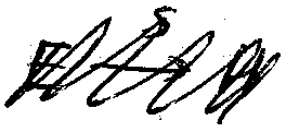
An entry immediately to the right of the upper H is crossed out. It may be another H.

Original source details: GELFAND-P000808

Trivial extension:

$$H = F \oplus E$$

Strong extension: (1) — is strong if $\exists$ a section [~~$E \xrightarrow{s} H$~~]



Uncertain reading. The crossed-out arrow is provisionally read as $E \xrightarrow{s} H$. Diagonal strokes cross the right-hand letter.

Original source details: GELFAND-P000808

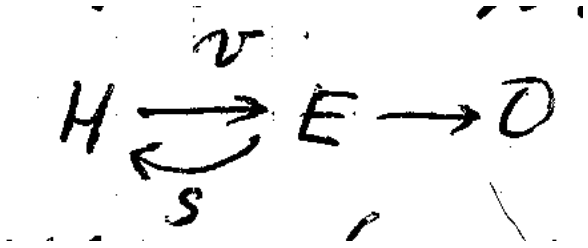
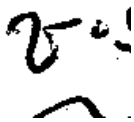


Diagram and typed description.

$$H \underset{s}{\overset{v}{\rightleftarrows}} E \longrightarrow 0$$

Original source details: GELFAND-P000808

$v \cdot s =$ id; s — a morphism of groups (and not of G-modules)



Original source details: GELFAND-P000808

Uncertain reading. The sign in $v \cdot s =$ id and over the arrow provisionally reads v.

[inserted below "groups": (abelian).]

(if s — a morph. of G-mods., the extension is trivial)

Set of equiv. classes of [inserted above the line: strong] extensions: $\mathrm{Ext}^1(E, F)$

Method: cohomology

$\mathrm{Hom}(E, F)$ — morph. of groups

this is a G-module. The action is given by

$$gu = g \cdot u \cdot g^{-1}$$

GELFAND-P000809

[source number: 43]

Def.

$$H^1(G, \mathrm{Hom}(E, F))$$

1-cocycles

$$f : G \longrightarrow \mathrm{Hom}(E, F)$$

$$f(gg') = f(g) + g \circ f(g') \circ g^{-1}. \tag{2}$$



Original source details: GELFAND-P000809

Uncertain reading. The lower loop of the first g after the plus sign in (2) is retraced. It reads g, with an indistinct adjoining stroke.

Z^1 — all cocycles.

1-coboundaries (B^1).

$$\exists\, u \in \mathrm{Hom}(E,F): \quad f(g) = g \circ u \circ g^{-1} - u$$

$$H^1 \underset{\text{def}}{=} Z^1/B^1.$$

There is a correspondence (one-to-one)

$$H^1 \cong \mathrm{Ext}^1(E,F)$$

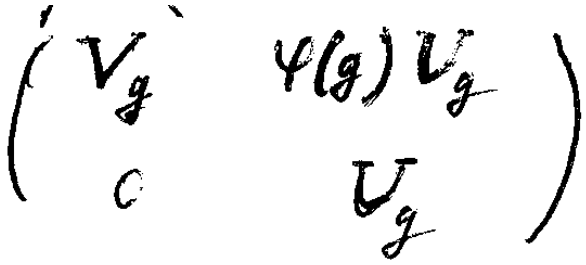
From the cocycle φ construct an extension:

$$H = F \oplus E$$

$$g(f,e) = (gf + \varphi(g)g \cdot e, \quad g \cdot e). \tag{3}$$

or in matrices:

$$\begin{pmatrix} V_g & \varphi(g)U_g \\ 0 & U_g \end{pmatrix}$$ — this is how the action is constructed.



Uncertain reading. The last letter at the top right is provisionally read as U_g, as on the diagonal. The lower-left sign is open at the right and provisionally reads 0.

Original source details: GELFAND-P000809

V_g, U_g — [...] reps. of G in the corresp. spaces.



Uncertain reading. An overwritten or crossed-out entry before “reps.” may be the numeral 2.

Original source details: GELFAND-P000809

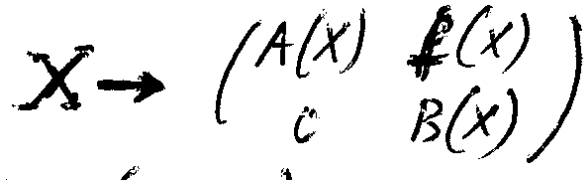
From this one can obtain the def. of a cocycle.

2 reps. equiv. $\Longleftrightarrow$ the cocycles are homologous

2. Extensions of modules over Lie algebras.

[on the right:]

$$X \longrightarrow \begin{pmatrix} A(X) & f(X) \\ 0 & B(X) \end{pmatrix}$$



Uncertain reading. The retraced letter at the top right provisionally reads f. The lower-left sign is open at the right and is rendered as 0.

Original source details: GELFAND-P000809

a cocycle is a linear map $f : \mathfrak{g} \longrightarrow \mathrm{Hom}(E,F)$

the def. of a cocycle has the form:

$$f([X,Y]) = X \circ f(Y) - f(Y) \circ X - Y \circ f(X) + f(X) \circ Y. \tag{4}$$

f — a coboundary if $\exists\, u$

$$f(X) = X \circ u - u \circ X. \tag{5}$$

Lemma 1. E, F have [inserted above the line: distinct] infinitesim. characters $\Rightarrow$

$$\Rightarrow \quad \mathrm{Ext}^1(E,F) = 0$$

$\mathfrak{g} \longmapsto U(\mathfrak{g})$ univ. enveloping algebra, $Z(\mathfrak{g})$ — its center

GELFAND-P000810

Let $u \in Z(\mathfrak{g})$ act in E by mult. by a scalar

$$u \cdot e = \chi(u) e$$

Then $\chi \in \mathrm{Hom}(Z(\mathfrak{g}), \mathbb{C})$

this is an infinitesim. character.

$H^1(\mathfrak{g}, \mathrm{Hom}(E,F))$ — is

$H^1(\mathfrak{g}, \mathfrak{h}, \mathrm{Hom}(E,F))$, $\mathfrak{h} \subset \mathfrak{g}$

[In the right margin: ....]



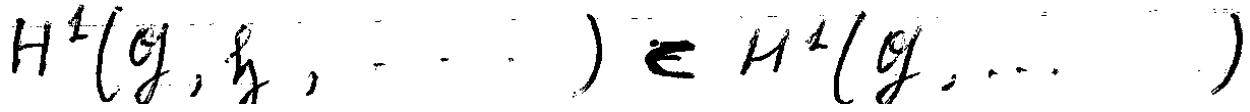
Uncertain reading. Isolated illegible mark

Original source details: GELFAND-P000810

(describes extensions trivial from the point of view of $\mathfrak{h}$).

$$Z^1(\mathfrak{g}, \mathfrak{h}, \mathrm{Hom}(E,F)) = \{f \in Z^1(\mathfrak{g}, \ldots), \quad f|_{\mathfrak{h}} = 0\}$$

$$H^1(\mathfrak{g}, \mathfrak{h}, \ldots) \in H^1(\mathfrak{g}, \ldots)$$



Original source details: GELFAND-P000810

Uncertain reading. The manuscript places the sign $\in$ between the groups.

Comments:

One can use the differential:

$$(df)(x,y) = f([x,y]) - xf(y) + yf(x)$$

$xf(y)$ — action Lie alg.

(above, written using the commutator)

3. Extensions of modules over Lie groups.

G-module — top. vect. space, G acts continuously.

$\mathfrak{g}$ — Lie alg.

$K \subset G$

|

max. comp. subgp.,

$\mathfrak{k}$ — corresp. alg.,

E, F — t.v.s.

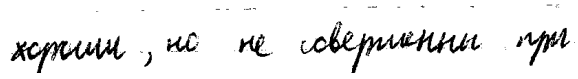
1st idea. cocycles

$$f : G \longrightarrow \mathrm{Hom}(E,F)$$

$(e,g) \longmapsto f(g)e$ cont. in g and e.

2nd idea: in place of E, F — the space of diff. vectors E_∞, F_∞

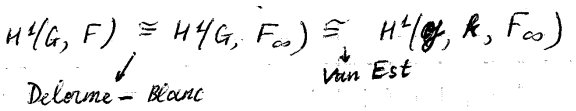
good, but imperfect when $E = \mathbb{C}$. Suppose so.



Original source details: GELFAND-P000810

Uncertain reading. The manuscript leaves the sense of "imperfect" unspecified.

Theorem:



Original source details: GELFAND-P000810

Diagram and typed description.

$H^1(G,F) \cong H^1(G,F_\infty) \cong H^1(\mathfrak{g},\mathfrak{k},F_\infty)$

↙ ↓

Delorme — Blanc Van Est



Original source details: GELFAND-P000810

Uncertain reading. The first algebra symbol in the right-hand group is retraced and provisionally reads $\mathfrak{g}$.

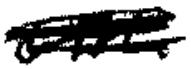
For $\mathbb{R}$ nontriv.

$$\begin{pmatrix} 1 & at \\ 0 & 1 \end{pmatrix}$$

— an extension of the triv. $\mathbb{R}$-module by the trivial one

there are no corresp. reps. of groups and algebras

(in cohom.: one must take cohom. triv. [crossed out: . . .] on the compact subgroup).



Undeciphered passage. word obscured by strokes

Original source details: GELFAND-P000810

GELFAND-P000811

[source number: 44]

Imperfection:
even if F — an irred. unit. rep., then it may be that

$$\dim H^1(G, F) \stackrel{?}{=} +\infty$$



Original source details: GELFAND-P000811

Uncertain reading. The sign has two nearly horizontal strokes with a superimposed angle. Its shape could be $\geq$. Equality agrees with the following example.

Example G — the aff. group of the line $ax + b$



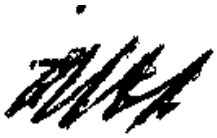
Original source details: GELFAND-P000811

Uncertain reading. The mark resembles R, with some strokes possibly crossing it out.

B^1 is everywhere dense in Z^1, but $\dim Z^1/B^1 = +\infty$

Reason: F weakly contains the trivial rep.

For semisimple groups all is well, [crossed out: . . .]



Original source details: GELFAND-P000811

Undeciphered passage. the original signs are illegible

Let G — semisimple;

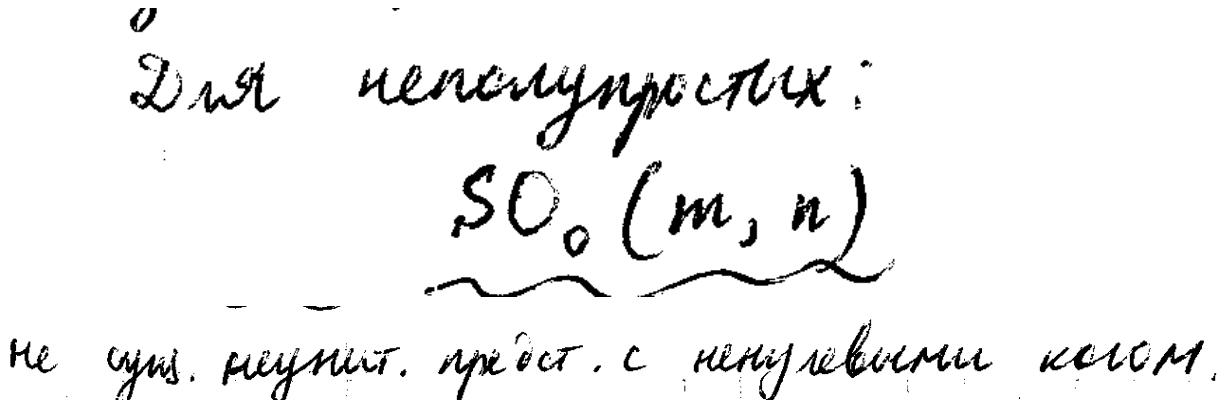
unit. irred. rep.

It is known when $H^1 = 0$.

For nonsemisimple groups:

$$SO_0(m,n)$$

there are no nonunit. reps. with nonzero cohom.



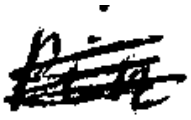
Original source details: GELFAND-P000811

Uncertain reading. The manuscript gives the heading “For nonsemisimple groups” and the abbreviation “nonunit.” alongside $SO_0(m,n)$.

Add. drawback of the 1st idea.

The lemma about infinitesim. char. is false!

There is a counterexample [crossed out: . . .] for $SL(2,\mathbb{R})$



Uncertain reading. Possibly “all”, with several letters obscured by transverse strokes.

Original source details: GELFAND-P000811

2nd idea: the lemma about infinitesim. chars. is true, but

[In the left margin, crossed out: . . .]



Uncertain reading. Possibly a repeated “but”.

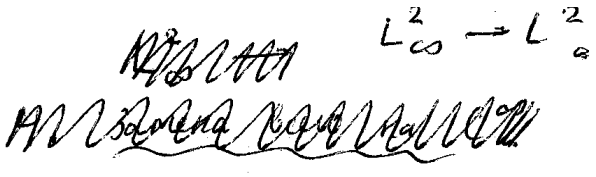
Original source details: GELFAND-P000811

$$H^1(G, \mathrm{Hom}(E_\infty, F_\infty))$$

is hard to compute.

$$L^2_\infty \longrightarrow L^2_\infty$$

[To the left and on the next line, crossed out: ~~$L^2_\infty \longrightarrow \ldots$~~; ~~Problem …~~]



Original source details: GELFAND-P000811

Undeciphered passage. The start of the upper entry provisionally reads L^2_∞ and an arrow is visible farther right. In the lower line, strokes obscure the words and signs following "Problem".

3rd idea.

$$E, F \rightsquigarrow E_K, F_K$$

(vectors incl. in finite-dimensional reps. of K).

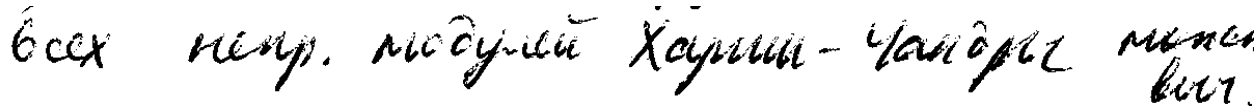


Original source details: GELFAND-P000811

Undeciphered passage. Two text fragments are crossed out. Isolated loops are visible, with the words obscured.

K-finite; K — max. comp. subgroup

$SL(2, \mathbb{R})$ — for all irred. Harish-Chandra modules one can compute



Original source details: GELFAND-P000811

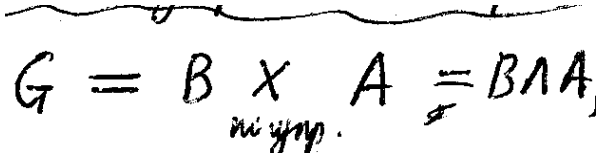
Uncertain reading. The abbreviation «непр.» is provisionally expanded as "irreducible".

$$H^1(\mathfrak{g}, \mathfrak{k}, \mathrm{Hom}(E_K, F_K))$$

algebra without topology.

4. The case of a semidirect product.

$$G = B \underset{\text{semidir.}}{\times} A = B \underset{\sigma}{\wedge} A, \qquad B = \mathbb{R}^n, \quad A \text{ — Lie group}$$



Original source details: GELFAND-P000811

Uncertain reading. The sign below the equality provisionally reads σ, possibly g. The sign between B and A is $\wedge$.

(e.g., the aff. group of the line).

B^* — dual space, $\langle b, x \rangle$ — duality.

$X = A$-orbit in B^*

(the action of A in B is precisely the def. of the semidirect prod.)

GELFAND-P000812

[To the left of the date, crossed out: . . ..]



Undeciphered passage. the word or digits are illegible

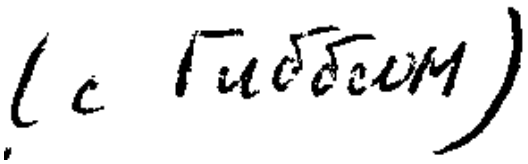
Original source details: GELFAND-P000812

12/II-79 Gelfand Sem.

What does one need to know? (according to A. A. Belavin):

I. Feynman.

1. Course for housewives, chs. 8, 9.
2. Statistical physics
3. Path integrals (with Gibbs)



Uncertain reading. The manuscript reads «Гиббсом» (Gibbs). The probable reference is to A. R. Hibbs, Feynman's coauthor on path integrals.

Original source details: GELFAND-P000812

II. Huang "Statist. Mechanics" Chapter: Onsager's solution.

III. F. Klein "Lectures on the History of Mathematics in the XIX Century"

[Crossed-out number: . . .]



Uncertain reading. possibly IV

Original source details: GELFAND-P000812

[4. The little books with blue cubes on the cover (against a white background) (Feynman: "Fund. Laws of Physics").]

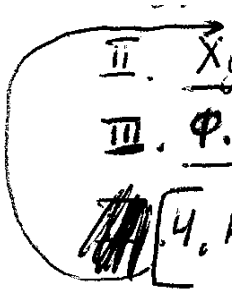
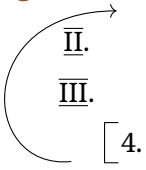


Diagram and typed description.



The arrow in the left margin runs from the crossed-out numeral to the space before II.

Original source details: GELFAND-P000812

IV. Dirac Article in the book “Dirac’s Monopole”

books. “Lectures on Quantum Mechanics”

“[ditto marks for ‘Lectures on’] Field Theory”

“General Relativity”

V. Akhiezer, Berestetskii “Quantum Electrodynamics”

(one should know the contents, but needn’t read it)

VI. Collection “Particles and Compensating Fields”, Article by Yang and Mills.

VII. Konopleva, Popov: “Gauge Fields” (the part written by Popov)

Belavin

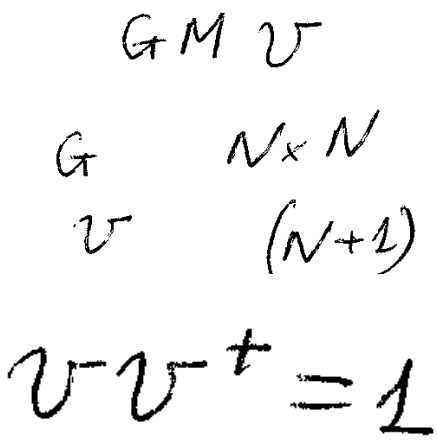
Problem put fwd. for a competition by the Chernogolovka Royal Phys. Soc.: or

What remains unsolv. about instantons?

A, B — 2 quaternionic matrices $N \times (N+1)$

One must find a parametrization [inserted above the line: “of such”] of pairs of matrices A and B, such that if $M = xA + B$, then MM^+ is real (for every quaternion x), up to the group action

$$\begin{array}{cc} & GMV \\ G & N \times N \\ V & (N+1) \times (N+1) \qquad VV^+ = 1 \end{array}$$



Uncertain reading. The same handwritten symbol is provisionally set as V throughout. Its shape could also be U or υ.

Original source details: GELFAND-P000812

There should be $8N - 3$ parameters in all.

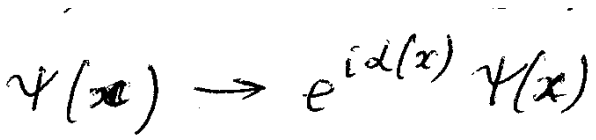
One must find a beautiful parametrization of such matrices.

On some exactly solvable 2-dimensional QFT problems.

A naive argument about non-escape. (Wilson)

$\psi(x)$ $\quad S[\psi, \ldots]$ — action, $\quad \delta S = 0$ — eqs. of motion

$\psi(x) \longrightarrow e^{i\alpha(x)}\psi(x)$ — gauge invariance



Uncertain reading. On the left of the formula, x is written over earlier strokes. The original sign is illegible.

Original source details: GELFAND-P000812

GELFAND-P000813

[source number: 63]

$$\langle \psi(x)\overline{\psi(y)}\rangle = e^{i(\alpha(x)-\alpha(y))}\langle \dots \rangle$$

$\Rightarrow$ correl. fns. $0 \Rightarrow$ impossibility [crossed out: ...] for charged objects.



Undeciphered passage. letters obscured by strokes

Original source details: GELFAND-P000813

Talk on hidden symmetry.

Passing from a cont. int. to a finite multiple integral can break the symmetry.

1) There is a theory

$$\Box\varphi = \sin\varphi, \qquad \varphi(x,t), \qquad \Box = \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2}$$

a quasiclass. spectrum has been obtained.

2) Next, there is a Ham. (Baxter):

$$H = \sum_{n=1}^{N}\sum_{\alpha=1}^{3} J_\alpha \sigma_n^\alpha \sigma_{n+1}^\alpha \tag{1}$$

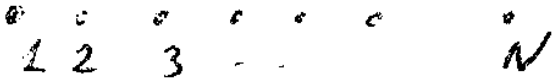


Diagram and typed description.

1 2 3 · · · N

Original source details: GELFAND-P000813

$$\sigma_n^\alpha = 1 \otimes 1 \otimes \dots \otimes \sigma^\alpha \otimes 1 \otimes \dots \otimes 1.$$
$$\alpha = 1,2,3$$



Uncertain reading. Below σ^α is a crossed-out mark, possibly n.

Original source details: GELFAND-P000813

[On the right] (1-dimensional quant. chain)

$\boxed{N+1=1}$ — periodicity condition.

$$\sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \qquad J_\alpha \in \mathbb{R}$$

H — a matrix $2^N \times 2^N$

Problem: find the eigenvecs. and eigenvalues.

In meaning, these models are analogous to 4-dim. gauge theories.

In the limit, 2) yields sin-Gordon.

Baxter solved syst. (1)

The Hamiltonian itself is Heisenberg's. (anisotropic, because of J_α — Heisenberg ferromagnet)

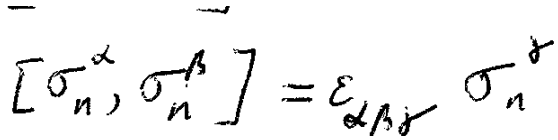
(or XYZ model)

Luther reduced (1) to the Thirring model.

The relations hold:

$$\begin{cases} [\sigma_n^\alpha, \sigma_m^\beta] = 0, & n \neq m \\ [\sigma_n^\alpha, \sigma_n^\beta] = \varepsilon_{\alpha\beta\gamma}\, \sigma_n^\gamma \\ [\sigma_n^1, \sigma_n^2]_+ = 0 \end{cases} \tag{2}$$

$[\,,\,]_+$ — anticommutator



Original source details: GELFAND-P000813

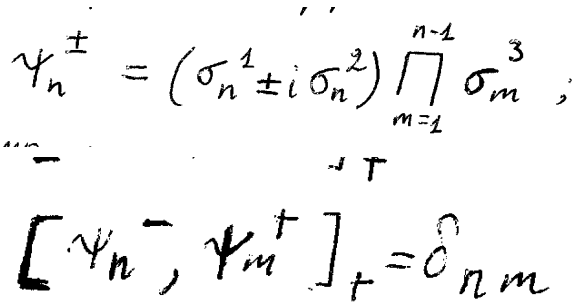
Uncertain reading. The middle line of (2) shows no factor before $\varepsilon_{\alpha\beta\gamma}$.

Let us make lattice fermions.

$$\psi_n^\pm = (\sigma_n^1 \pm i\sigma_n^2) \prod_{m=1}^{n-1} \sigma_m^3; \tag{3}$$

these are fermions, i.e.

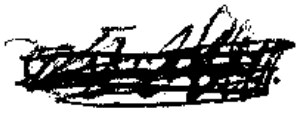
$$\begin{cases} [\psi_n^+, \psi_m^+]_+ = 0 \\ [\psi_n^-, \psi_m^+]_+ = \delta_{nm} \end{cases} \tag{4}$$



Original source details: GELFAND-P000813

Uncertain reading. In (3) there is no factor before the parentheses. In (4) the right-hand side is δ_{nm}.

In [crossed out: . . .] fact — this is how Onsager solved the Ising model:



Undeciphered passage. the word is obscured by strokes

Original source details: GELFAND-P000813

express σ in terms of ψ: H will become quadr. and everything is diagonalized

GELFAND-P000814

Let us take a Fourier transf.

$$\widetilde{\psi}_k^{\pm} = \sum_n \psi_n^{\pm} e^{ikn} \tag{5}$$

[crossed out: . . .]



Uncertain reading. Possibly “I will formulate” or “we will formulate”. Cancellation strokes obscure the ending.

Original source details: GELFAND-P000814

Assertion.

$$H = \sum_n (\sigma_n^1 \sigma_{n+1}^1 + \Delta \sigma_n^3), \qquad \Delta \text{ — a number} \tag{6}$$

$$\dot{\sigma}_n(t) = i[\sigma_n, H] \tag{7}$$

— eqs. of motion; for the solutions $\forall t$ the commut. rels. hold (2)

$$\sigma_n(0) = \sigma_n.$$

After substitution the eqs. become linear (in ψ)

$\Rightarrow$ this is equiv. to finding eigenvectors.

(σ can be expressed in terms of ψ)

We obtain the eqs.

$$\dot{\psi}_n = \psi_{n+1} - \psi_{n-1} \tag{8}$$

Expr. for the Hamiltonian in the orig. problem.

$$H \sim \sum_n \left(\psi_n \psi_{n+1}^+ - \psi_n^+ \psi_{n+1} + \psi_n^+ \psi_n \psi_{n+1}^+ \psi_{n+1}\right) \tag{9}$$

[crossed out: ~~σ_1 and σ_2~~.]



Uncertain reading. The cancelled symbols provisionally read σ_1 and σ_2. The indices may be superscripts.

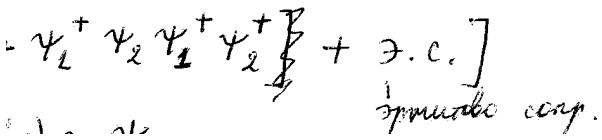
Original source details: GELFAND-P000814

σ_n is expr. in terms of ψ by induction.

(9) — Ham. of the lattice Thirring model (already nonlin.)

Passing to the continuous case, we obtain:

$$\int dx \left[\psi_1^+(x) \frac{\partial \psi_2(x)}{\partial x} + \psi_1^+ \psi_2 \psi_1^+ \psi_2^+ \right] + \underset{\text{Hermitian conj.}}{\text{h.c.}} \tag{10}$$



Uncertain reading. The subscript on the third factor is retraced and reads 1. The first closing bracket is crossed out.

Original source details: GELFAND-P000814

$$x = na, \qquad \psi(x) \approx \psi_n$$

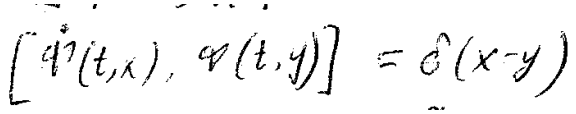
ψ_1, ψ_2 — from different even and odd lattice sites

1975 Mandelstam and...: quant. Thirring model $\sim$ quant. mod. sin-Gordon

quant. mod. sin-Gord

$$\Box\varphi = \sin\varphi \tag{11}$$

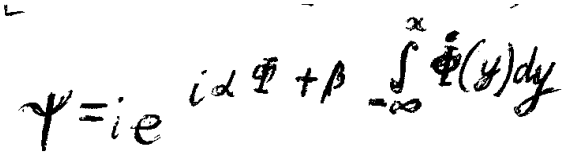
$$[\dot{\varphi}(t,x), \varphi(t,y)] = \delta(x-y) \tag{12}$$



Uncertain reading. The right-hand side of (12) shows $\delta(x-y)$ without a factor i.

Original source details: GELFAND-P000814

$$\psi = ie^{i\alpha\Phi + \beta \int_{-\infty}^{x} \dot{\Phi}(y)\, dy} \tag{13}$$



Uncertain reading. The field in (13) is set as Φ, possibly φ. There is an i before e and no additional i is visible before β.

Original source details: GELFAND-P000814

$\Downarrow$ (when $\alpha\beta = 4\pi$)

$$[\psi(x), \psi^+(y)]_+ = \delta(x-y) \tag{14}$$

End of the introduction.

Sasha Zamolodchikov. Zamolodchikov algebra.

[Under the name:] (Moscow)

$$\underset{\text{symbols}}{A_i(\theta)}\,, \qquad 1 \le i < M$$

GELFAND-P000815

[source number: 64]

[At the upper left: . . .]



Uncertain reading. An isolated short vertical stroke with a rightward hook at its top. Its meaning is unclear.

Original source details: GELFAND-P000815

One can write products

$$A_i(\theta_1)A_\alpha(\theta_2) = S_{ik}^{\alpha\beta}(\theta_1 - \theta_2)A_\beta(\theta_2)A_k(\theta_1) \tag{15}$$

$$1 \le \beta < M, \qquad 1 \le \alpha < M.$$



Uncertain reading. Both upper bounds in the manuscript use strict $<$. The example below uses A_2 with $M = 2$.

Original source details: GELFAND-P000815

[crossed out: . . .]



Undeciphered passage. the original mark is illegible

Original source details: GELFAND-P000815

$M = 2$

$$\begin{cases} A_1 A_1 = a A_1 A_1 + d A_2 A_2 \\ A_1 A_2 = b A_2 A_1 + c A_1 A_2 \end{cases} \tag{16}$$

and 2 more rels. obt. by replacing 1 with 2.

This still has to be made compatible with associativity.

On S the corresp. condition is imposed

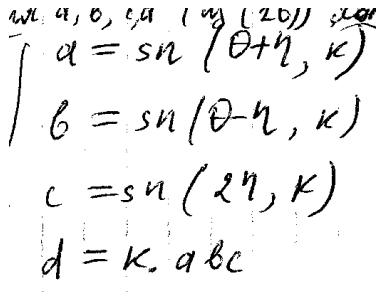
$$\big(A_1(\theta_1) A_1(\theta_2)\big) A_2(\theta_3) = A_1(\theta_1) \big(A_1(\theta_2) A_2(\theta_3)\big)$$

The eqs. lead to factorization of multiparticle S-matrices

Thus he found the S-matrix of sin-Gordon etc.

The eqs. for a, b, c, d (from (16)) are solved explicitly (!)

$$\begin{cases} a = \mathrm{sn}(\theta + \eta, k) \\ b = \mathrm{sn}(\theta - \eta, k) \\ c = \mathrm{sn}(2\eta, k) \\ d = k \cdot abc \end{cases} \tag{17}$$



Uncertain reading. The second argument of sn and the factor in d are provisionally set as k, possibly κ.

Original source details: GELFAND-P000815

These are all solns. up to a factor.

Baxter used such objects

$$T^{\alpha_1, \beta_1, \ldots, \alpha_N, \beta_N}(\theta) = S^{\alpha_1 \beta_1}_{i_1 i_2} S^{\alpha_2 \beta_2}_{i_2 i_3} \ldots S^{\alpha_N \beta_N}_{i_N i_1} \tag{18}$$

The result is a $2^N \times 2^N$ matrix. It turns out that

[To the left of (19) crossed out: . . .]



Undeciphered passage. the original mark is illegible

Original source details: GELFAND-P000815

$$[T(\theta), H] = 0 \tag{19}$$

(⇒ infin. set of integrals of motion)

$$[T(\theta_1), T(\theta_2)] = 0 \tag{20}$$

(⇒ in involution)

In fact he solved it exactly.

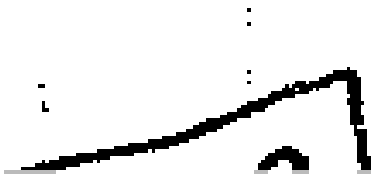


Diagram and typed description.



Angular mark at the lower left. Its strokes end at the same horizontal boundary in the source image. The full shape and purpose are unclear.

Original source details: GELFAND-P000815

GELFAND-P000816

19/II-79 Gelfand Sem.

Belavin.

$$A_i(\theta), \qquad 1 \le i \le M, \qquad \text{— symbols}$$

θ — a parameter

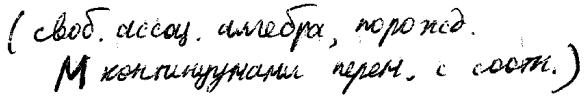
Zamolodchikov algebra.

$$A_i(\theta_1)A_\alpha(\theta_2) = S_{ik}^{\alpha\beta}(\theta_1 - \theta_2)A_\beta(\theta_2)A_k(\theta_1) \tag{1}$$

Associativity is required too. Then cubic rels. arise.

$$A_i(\theta_1)A_j(\theta_2)A_k(\theta_3)$$

[On the right:] (free assoc. algebra gen. by M continua of vars. with rels.)



Uncertain reading. The marginal phrase reads «континуумами перем.» (continua of vars.).

Original source details: GELFAND-P000816

Interpretation (physical)



Diagram and typed description.



Original source details: GELFAND-P000816

pos. — particles, θ_i — velocities



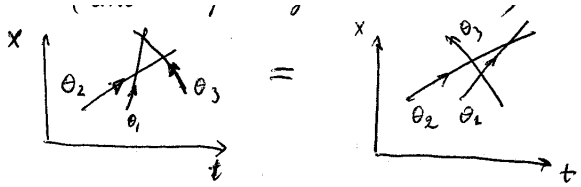
Original source details: GELFAND-P000816

Uncertain reading. The abbreviation below the drawing provisionally reads «полож.» (positions).

$A_i A_\alpha$ — the ith part. is farther left.

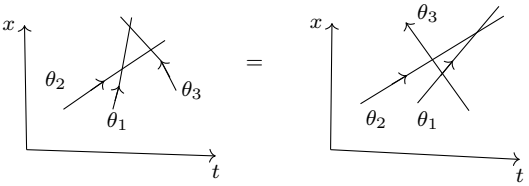
$\theta_1 > \theta_2 \Rightarrow$ the ith catches up.

They can catch up in different orders, but the beginning and end matter (what happens in between does not matter)



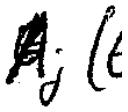
Original source details: GELFAND-P000816

Diagram and typed description.



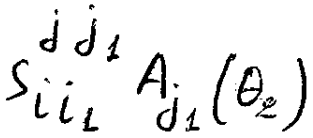
The arrowhead on the descending θ_3 line on the left is retraced. Its direction is provisionally drawn up and to the left.

$$
\begin{aligned}
&\big(A_i(\theta_1)A_j(\theta_2)\big)A_k(\theta_3)\\
&= S^{jj_1}_{ii_1}A_{j_1}(\theta_2)\big(A_{i_1}(\theta_1)A_k(\theta_3)\big) =\\
&= S^{jj_1}_{ii_1}(\theta_{12})S^{kk_1}_{i_1i_2}(\theta_{13})\big(A_{j_1}(\theta_2)A_{k_1}(\theta_3)\big)A_{i_2}(\theta_1) =\\
&= S^{jj_1}_{ii_1}(\theta_{12})S^{kk_1}_{i_1i_2}(\theta_{13})S^{k_1k_2}_{j_1j_2}(\theta_{23})\\
&\quad A_{k_2}(\theta_3)A_{j_2}(\theta_2)A_i(\theta)
\end{aligned}
$$



Original source details: GELFAND-P000816

Uncertain reading. The letter A in $A_j(\theta_2)$ in the first product is retraced. The original stroke is illegible.



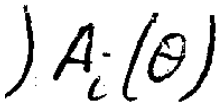
Original source details: GELFAND-P000816

Uncertain reading. There is no written argument for $S^{jj_1}_{ii_1}$ in the first equality.



Original source details: GELFAND-P000816

Uncertain reading. The θ in θ_3 after A_{k_2} in the final product is retraced. The original stroke is illegible.



Uncertain reading. The last factor is written $A_i(\theta)$ without further numerical indices.

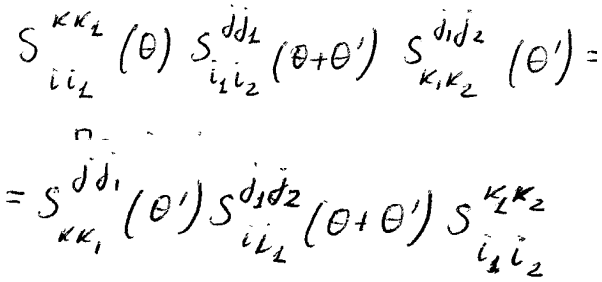
Original source details: GELFAND-P000816

GELFAND-P000817

[source number: 67]

The eqs. take the form:

$$S_{ii_1}^{kk_1}(\theta)S_{i_1i_2}^{jj_1}(\theta+\theta')S_{k_1k_2}^{j_1j_2}(\theta') = S_{kk_1}^{jj_1}(\theta')S_{ii_1}^{j_1j_2}(\theta+\theta')S_{i_1i_2}^{k_1k_2} \tag{2}$$



Uncertain reading. The indices in (2) follow the manuscript. The last S has no written argument.

Original source details: GELFAND-P000817

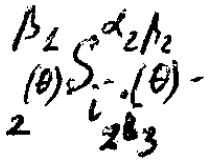
Over i_1, j_1, k_1 sum.

In all: M^6 eqs. for M^4 fns. $S_{ik}^{\alpha\beta}(\theta)$

S plays the role of an S-matrix.

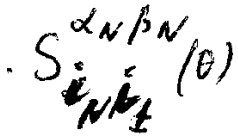
Put

$$T_{\vec{\alpha},\vec{\beta}}(\theta) = S_{i_1i_2}^{\alpha_1\beta_1}(\theta)S_{i_2i_3}^{\alpha_2\beta_2}(\theta)\cdot\ \cdot\ \cdot S_{i_Ni_1}^{\alpha_N\beta_N}(\theta) \tag{3}$$



Uncertain reading. The retraced lower indices of the middle factor provisionally read i_2i_3.

Original source details: GELFAND-P000817



Uncertain reading. The retraced lower indices of the last factor provisionally read i_Ni_1, possibly i_Nk_1.

Original source details: GELFAND-P000817

The result will be a matrix $M^N \times M^N$ (“transfer matrix”)

[On the right:] $(\alpha_1, \ldots, \alpha_N)$ — indices.

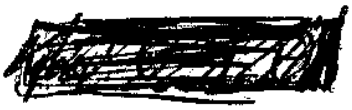
$(\beta_1, \ldots, \beta_N)$ — 2nd set of inds.

It is asserted that

$$[T(\theta), T(\theta')] = 0 \tag{4}$$

[To the right of zero, crossed out: . . .]

Undeciphered passage. the entry is illegible



Original source details: GELFAND-P000817

(by virtue of the factorization eqs.).

[On the left, an underlined entry is crossed out: . . .]

Undeciphered passage. the entry is illegible



Original source details: GELFAND-P000817

Factoriz. of the S-matr. is a consequence of a fin. number of conserv. laws.

Explicit form of the eqs. for $M = 2$

(case of Maxwell’s eqs.)

$$\begin{cases} A_1 A_1 = a\, A_1 A_1 + d\, A_2 A_2 \\ A_1 A_2 = b\, A_2 A_1 + c\, A_1 A_2 \end{cases} \qquad \begin{pmatrix} A_1(\theta_1),\ A_2(\theta_2) \\ a(\theta_1 - \theta_2) \end{pmatrix} \tag{5}$$

[On the right:] and 2 more eqs. with 1 replaced by 2 (with the same coeffs.) (see $(5)_2$)

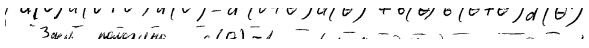
There are 4 fns. a, b, c, d.

There could be 16 fns. In fact 5 or 6 fns. (from the eqs.).

Here 4 are retained

The factorization eqs. give:

$$\begin{cases} a(\theta)a(\theta+\theta')d(\theta') = d(\theta+\theta')a(\theta') \\ \qquad\qquad\qquad\qquad + b(\theta)b(\theta+\theta')d(\theta') \end{cases} \tag{6}$$



Original source details: GELFAND-P000817

Diagram and typed description.

A row of short strokes beneath the written line of (6).

Here we have set $c(\theta) \equiv 1$ (the eqs. are homogeneous — let us divide).



Uncertain reading. The verb provisionally reads «положено» (has been set), possibly «положим» (let us set).

Original source details: GELFAND-P000817

[On the right:] (6 funct. eqs. in all)

$$\begin{cases} A_2A_2 = a\,A_2A_2 + d\,A_1A_1 \\ A_2A_1 = b\,A_1A_2 + c\,A_2A_1 \end{cases} \qquad (5)_2$$

Uncertain reading. The original mark under the retraced numeral 5 in $(5)_2$ is illegible.



Original source details: GELFAND-P000817

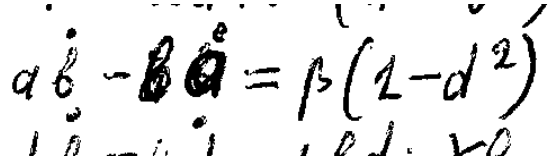
$$S_{ik}^{\alpha\beta}(0) = \delta_{i\beta}\delta_{\alpha k} \qquad (7)$$

— init. condition

We obtain diff. eqs. from (6):

$$\begin{cases} \dot a = \alpha a - \beta b + \gamma ad & \\ \dot d = \alpha d + \gamma(a^2 - b^2) & \dot a(0) = \alpha \\ a\dot b - b\dot a = \beta(1 - d^2) & \dot b(0) = \beta \\ d\dot b - b\dot d = \alpha bd + \gamma b - \beta ad & \dot d(0) = \gamma \\ \text{and 2 more eqs.} & \end{cases} \qquad (8)$$

Uncertain reading. The $b\dot a$ in the third line of (8) is retraced. The original strokes are illegible.



Original source details: GELFAND-P000817

GELFAND-P000818

All the eqs. are independent.

3 algebr. relations arise, but one follows from the other 2.

Elliptic fns. are obtained (Baxter)

A noncommut., nonassoc. algebra arises, of quadruples of fns.

Uncertain reading. The word reads «четверок» (quadruples). Strokes above «е» may indicate «ё».



Original source details: GELFAND-P000818

It would be interesting to find an explicit realiz. (e.g., on matrices)

Yu. I. Manin's idea: take points on an ell. curve, of finite order

Let us expand

$$T(\theta) = \sum_{\ell=0}^{\infty} I_\ell \theta^\ell \tag{9}$$

Then

$$[I_n, I_m] = 0 \tag{10}$$

and in the example ($M = 2$):

$$I_0^{-1}\,\cancel{(0)}\,I_1\,\cancel{(0)} = \sum_{n=1}^{N}\sum_{\alpha=1}^{3} J_\alpha \sigma_n^\alpha \sigma_{n+1}^\alpha \quad (= T^{-1}(0)\dot{T}(0)) = H \tag{11}$$



Original source details: GELFAND-P000818

Uncertain reading. A mark after I_0^{-1} is crossed out. It may read (0).



Original source details: GELFAND-P000818

Uncertain reading. A separate mark after I_1 is crossed out. It too may read (0).

Here

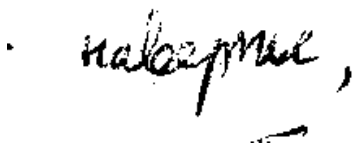
$$\sigma_n^\alpha = 1 \otimes \ldots \otimes \sigma^\alpha \otimes \ldots \otimes 1, \qquad \sigma^\alpha \text{ — the Pauli matrices.}$$

$$[I_\ell, H] = 0, \quad \ell = 0, 1, 2, \ldots \tag{12}$$

The transfer matr. is written in terms of S, i.e., in terms of ellipt. fns.

I_2 contains 3 spins, and so on.

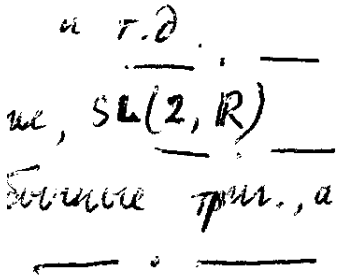
I. M. G. a, b, c, d — probably, $SL(2, R)$



Original source details: GELFAND-P000818

Uncertain reading. The word provisionally reads «наверное», possibly «наверно» (probably).

$d = 0 \Rightarrow$ we obtain ordinary trig. fns. instead of ellipt. fns.



Original source details: GELFAND-P000818

Diagram and typed description.

Three rows of separate strokes beneath the entries about spins, $SL(2, R)$, and trig. fns.

General assertion (now M is arbitrary):

Put

$$\dddot{\overline{H}} = \sum_{n=1}^{N} \dot{S}^{\alpha\beta}_{ik}(0)\sigma^{i\beta}_{n}\sigma^{\alpha k}_{n+1} = I_0^{-1} I_1 \qquad \text{(see (9))}. \tag{13}$$



Undeciphered passage. A mark above H is crossed out. The original sign is illegible.

Original source details: GELFAND-P000818

σ — a matrix of size $M \times M$.

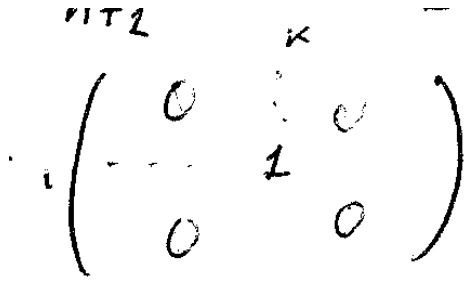


Diagram and typed description.

$$\sigma^{ik} = i \begin{pmatrix} & & k & \\ & 0 & \vdots & 0 \\ \cdots & & 1 & \\ & 0 & & 0 \end{pmatrix}$$

basis in the space of matrices (1 in position i, k).

Original source details: GELFAND-P000818

$$\sigma^{ik}_{n} = 1 \otimes \ldots \otimes \sigma^{ik}_{n} \otimes \ldots \otimes 1$$

We have a loc. Hamiltonian.



Uncertain reading. The abbreviation before "Hamiltonian" provisionally reads «лок.» (loc.).

Original source details: GELFAND-P000818

Zamolodchikov obtained particular solutions (for different M)

Let us use them for $M = 3$, We obtain

$$H = \sum_{n} \dddot{(}(\vec{S}_n \vec{S}_{n+1})^2 - 2\vec{S}_n \vec{S}_{n+1} \tag{14}$$



Undeciphered passage. An entry between the summation sign and the opening parenthesis is crossed out. The original signs are illegible.

Original source details: GELFAND-P000818

$M = 2$ — isotr. Heisenberg ferromagnet

$$H = \sum_{n} \vec{\sigma}_n \vec{\sigma}_{n+1} \tag{15}$$

spin $1/2$ — there is nothing else inv.

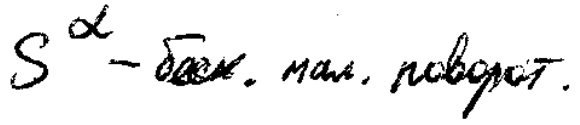
GELFAND-P000819

[source number: 68]

In (14): S^α are such that

$$[S^\alpha, S^\beta] = \varepsilon_{\alpha\beta\gamma} S^\gamma, \tag{16}$$

S^α — inf. small rotation.



Uncertain reading. The first abbreviation before “small rotation” provisionally reads «беск.» (inf.). Some strokes overlap.

Original source details: GELFAND-P000819

(rep. of the group $SO(3)$)

adj. = standard

There are no parameters here

$M = 3 \Rightarrow$ there is no general soln.

(!) It is unknown how to add parameters to Ham. (14) so as to fit Zamolodchikov's scheme.

This scheme cont. all fully integr. quant. systems (2-dimensional)

($\Rightarrow$ classical ones too after taking the quasiclass. limit).

$$H = \sum_{n=1}^{N} \dot{S}_{ik}^{\alpha\beta}(0) \sigma_n^{i\beta} \sigma_{n+1}^{\alpha k}; \tag{17}$$

I.M.G.'s question: rels. on S — of Maurer–Cartan type

(17) — the most general form of Ham. with nearest-neighbor interact.

It always has a nontriv. integral, since the diff. eqs. extend to the 2nd neighborhood.

$$S = S(0) + S'(0)\theta + \frac{S''(0)\theta^2}{2}$$

S'' is recovered.

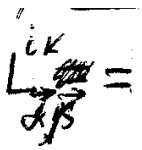
$A_i(\theta)$ are matrices $\Rightarrow$ there is assoc. automatically

There is a way to multiply algebras.

Suppose we found $S_{ik}^{\alpha\beta}$. Let us construct

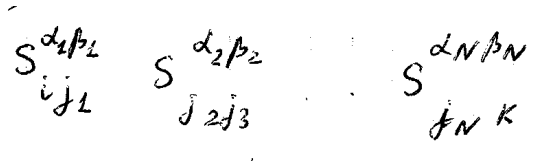
(e.g. $M = 2$)

$$L_{\vec{\alpha}\vec{\beta}}^{ik} = S_{ij_1}^{\alpha_1\beta_1} S_{j_2j_3}^{\alpha_2\beta_2} \ldots S_{j_Nk}^{\alpha_N\beta_N} \quad \text{(we did not take the trace)} \tag{18}$$



Uncertain reading. The lower indices of L are set as $\vec{\alpha}\vec{\beta}$. There is a crossed-out entry alongside them.

Original source details: GELFAND-P000819



Uncertain reading. The second factor has lower indices $j_2 j_3$, while the first ends with j_1. Two dots separate the second and last factors.

Original source details: GELFAND-P000819

We get the matrix

$$L = \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{pmatrix} = (L^{\alpha\beta}) \tag{19}$$

its elems. — are matrices too.

[crossed out: ~~...~~] Rels. of type (2) $--$ hold for the matrix L

Undeciphered passage. The crossed-out sign before "Rels." is illegible.



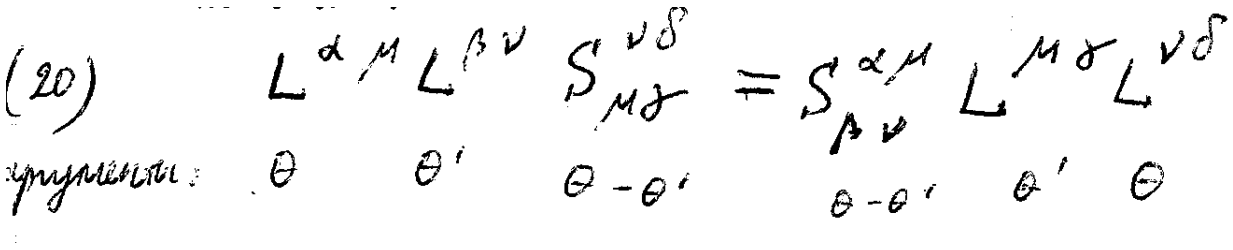
Original source details: GELFAND-P000819

$$S^{\alpha\mu}_{ik} S^{\beta\nu}_{kj} S^{\nu\delta}_{\mu\gamma} = S^{\alpha\mu}_{\beta\nu} S^{\mu\gamma}_{ik} S^{\nu\delta}_{kj} \tag{2}'$$

L^{ik} — matr. $M^N \times M^N$ for all i, k.

There are relations:

$$\begin{array}{lccccccc} & L^{\alpha\mu} & L^{\beta\nu} & S^{\nu\delta}_{\mu\gamma} & = & S^{\alpha\mu}_{\beta\nu} & L^{\mu\gamma} & L^{\nu\delta}. \\ \text{arguments:} & \theta & \theta' & \theta-\theta' & & \theta-\theta' & \theta' & \theta \end{array} \tag{20}$$



Original source details: GELFAND-P000819

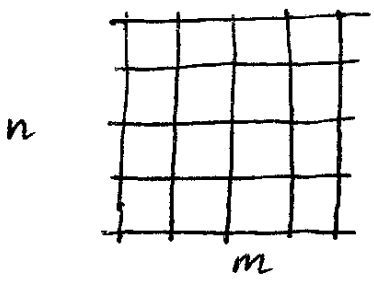
Uncertain reading. The arguments are written under their corresponding factors. On the right, the two L factors are marked first θ', then θ.

GELFAND-P000820

Now suppose

$$A_{\alpha\mu} = L^{\alpha\mu} \tag{21}$$

We get a new algebra of higher order



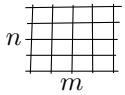
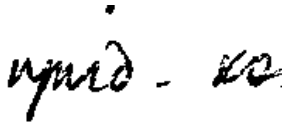


Diagram and typed description. 2-dim. lattice rolled into a torus.

Original source details: GELFAND-P000820

Onsager in ’44 — 2-dim. Ising model

—| |— ’65 — devis. commut. transfer matrix.



Uncertain reading. The abbreviation before “commut.” provisionally reads «прид.» (devis.). «пред.» (propos.) is also possible.

Original source details: GELFAND-P000820

$$\sigma_{n,m} = \pm 1$$

We need to calc. the sum:

$$Z = \sum_{\{\sigma_{n,m}\}} e^{J_1 \sum \sigma_{n,m}\sigma_{n,m+1} + J_2 \sum \sigma_{n,m}\sigma_{n+1,m} \cdots} \tag{22}$$



Undeciphered passage. The entry at the end of the exponent is crossed out. Its signs are illegible.

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(nearest-neighbor interact.)

Idea:

$$Z = \operatorname{Tr} T^N \tag{23}$$

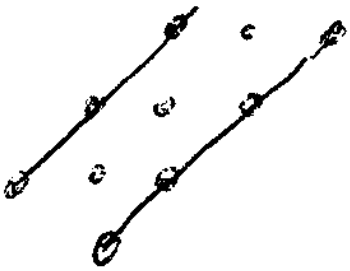
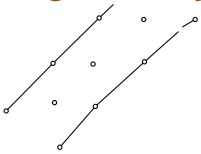


Diagram and typed description.



The strokes meet at the top circle on the left line. A separate arrowhead cannot be distinguished.

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[At right:] Onsager: we go along the diagonals (we sum over the spins on the interm. diag.)

$$T_{\vec{\alpha}\,\vec{\beta}} = S_1 \dots S_n, \tag{24}$$

where



Diagram and typed description.

(25) $S_{ik}^{\alpha\beta} =$ 

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We get the tr. matr. $T(J_1, J_2)$

Prop.

$T(J_1, J_2)$ commut. wh



Undeciphered passage. The last visible letters are «пр», possibly the start of «при» (“when”). The entry breaks off.

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